

ARIN 5201 & MAIE 5421

Machine Learning for Computer Vision

Wenhan LUO

9/4/2026

Course Components

- Machine Learning and Computer Vision Basics
- Advance Topics in Computer Vision

Assessment Scheme

No.	Assessment Method	Weighting (%)
1	Attendance	10
2	Assignments	30
3	Final Project Proposal	10
4	Final Project Presentation	20
5	Final Project Technical Report	30

- Three assignments are released in the 4th, 7th, 10th week. The answers are expected to be returned by the end of the 7th, 10th, 13th week. **Late submissions will not be accepted under normal circumstances.** Should you encounter any issues that prevent you from submitting on time or completing an assignment, you must communicate with the TAs well in advance of the deadline.
- The final project proposal should be submitted by the end of the 7th week. The technical report should be submitted by the end of the 13th week.

Content

Topic	Week
Machine Learning and Computer Vision Basics	
Introduction to Computer Vision	1
Machine Learning Elements	2
Deep Learning	3-4
 Advanced Topics in Computer Vision	
Image Classification with CNNs	5-6
Detection & Segmentation	7
Attention Mechanism & Transformer	8
Generative Models	9-10
Multi-modal Learning	11
Final project representation	12-13

Course Logistics

Lectures: Friday, 7:00 - 9:50 pm, G012, LSK Building

Assignments/announcements

Canvas

Office Hour

10:00 AM - 11:00 AM, Tuesday, Room 5321 (lift 3)

Instructor and TAs

Instructor

Wenhan LUO, whluo@ust.hk

TAs

- Zhizhou ZHONG, zzhongax@connect.ust.hk
- Bo YANG, byangbf@connect.ust.hk

Resources and Materials

1. David Forsyth and Jean Ponce. "Computer Vision: A Modern Approach", Prentice Hall, 2003.
2. Ian Goodfellow, Yoshua Bengio, Aaron Courville. "Deep Learning". MIT Press, 2016.
3. Christopher Bishop, "Pattern Recognition and Machine Learning", Springer New York, 2006.
4. Aston Zhang, Zachary C. Lipton, Mu Li, Alexander J. Smola, "Dive into Deep Learning", Cambridge University Press, 2023.

Introduction to Computer Vision

Introduction to Computer Vision (CV)

- What is Computer Vision
- Brief History of Computer Vision
- Key Research Areas of Computer Vision
- Applications of Computer Vision
- Why Study Computer Vision

What a person sees



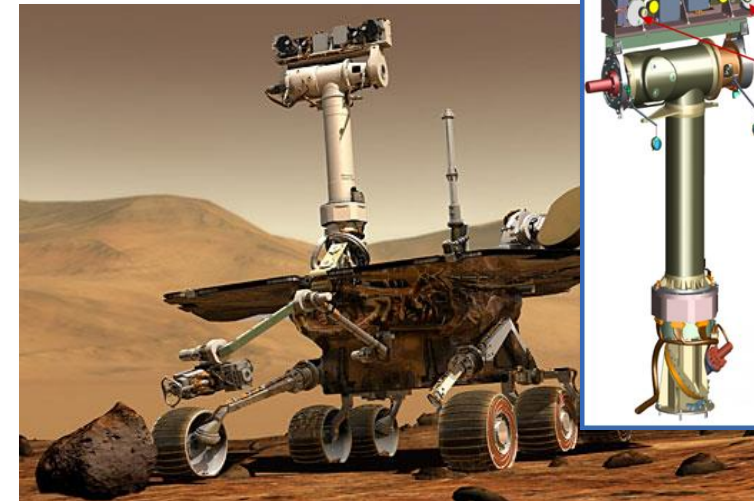
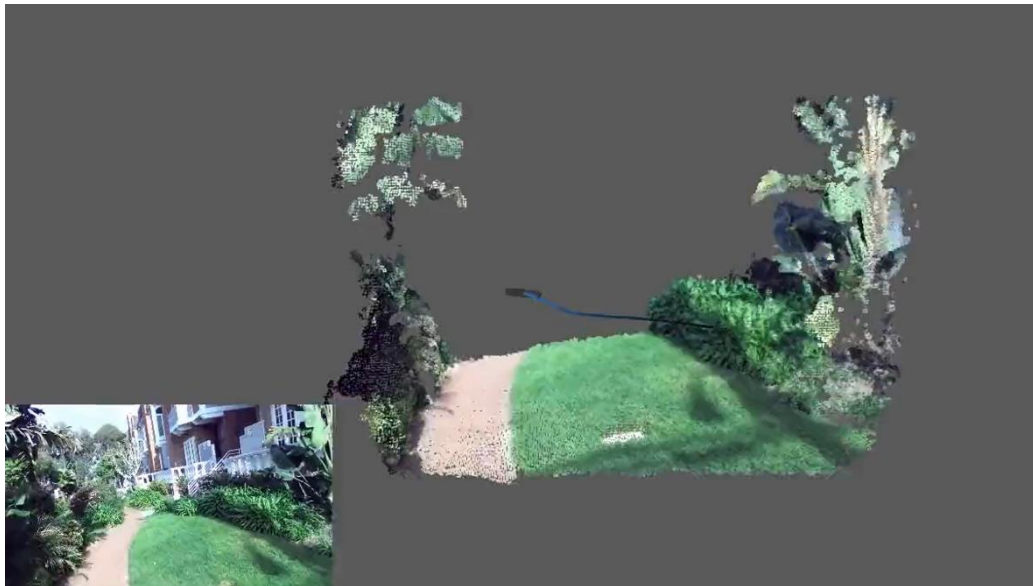
Photo by Svetlana Lazebnik

The goal of Computer Vision is to
give computers
(super) human-level intelligence

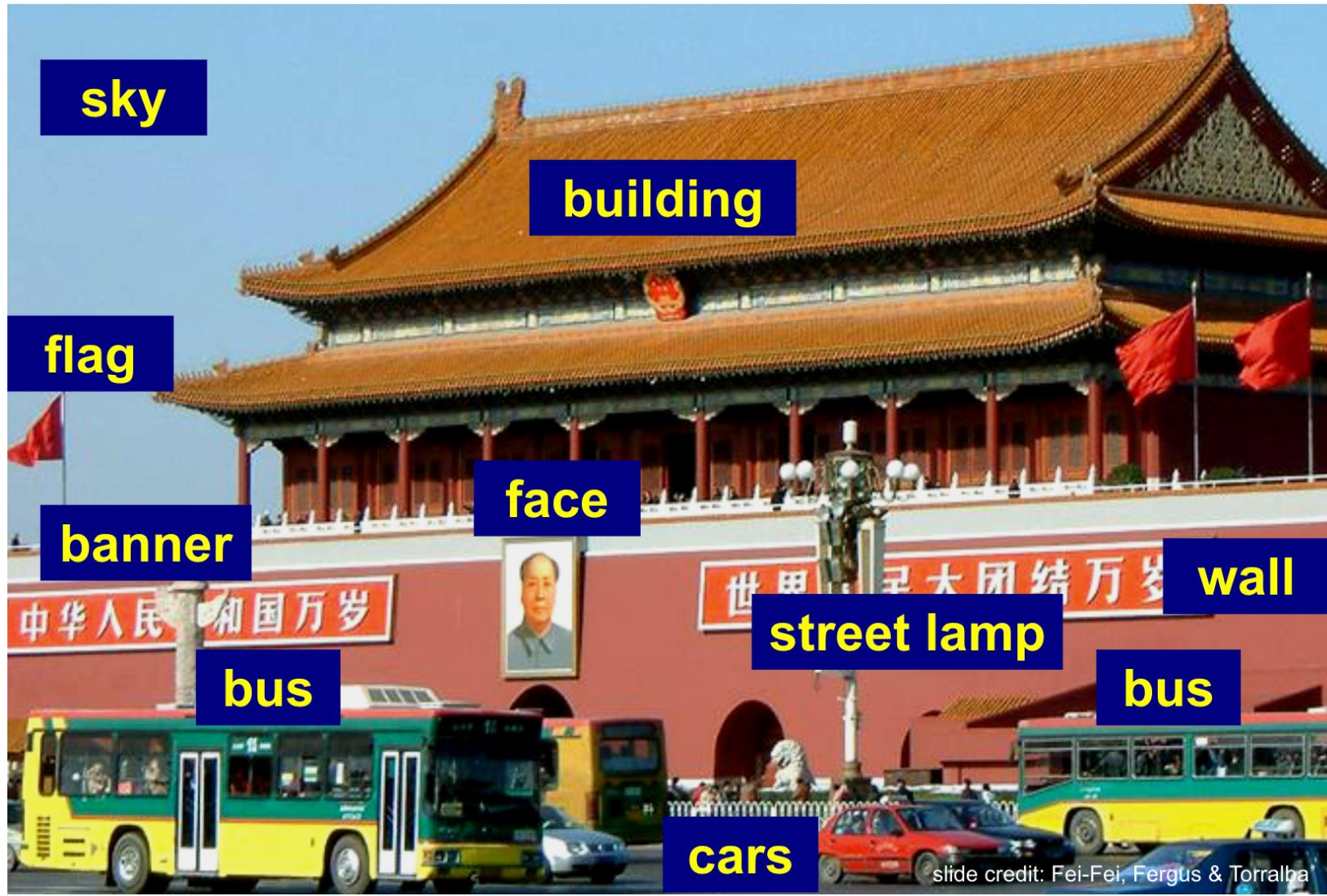
Compute the 3D shape of the world



ZED 2i Camera



Recognize objects and people



Improve photos (“Computational Photography”)



Super-resolution (source: 2d3)



Low-light photography

9/4/2026 (credit: [Hasinoff et al., SIGGRAPH ASIA 2016](#))



Depth of field on cell phone camera
(source: [Google Research Blog](#))

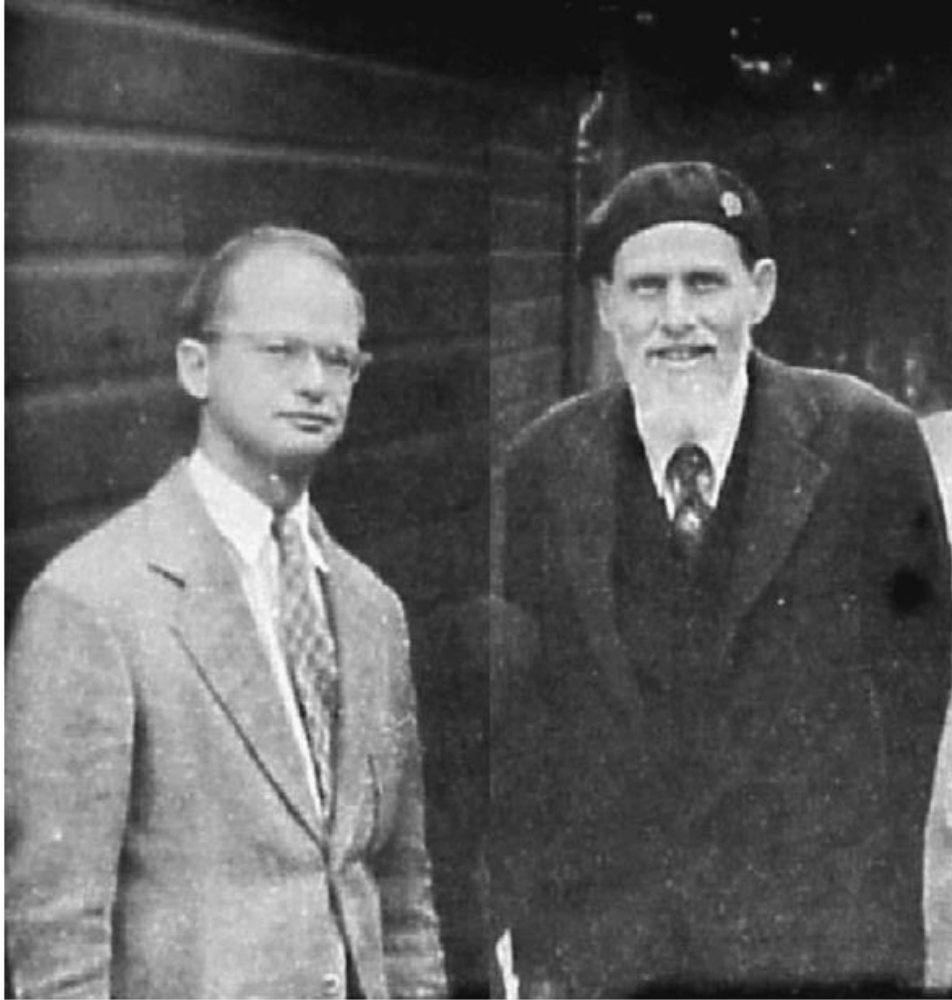


Removing objects
([Google Magic Eraser](#))

Introduction to Computer Vision (CV)

- What is Computer Vision
- **Brief History of Computer Vision**
- Key Research Areas of Computer Vision
- Applications of Computer Vision
- Why Study Computer Vision

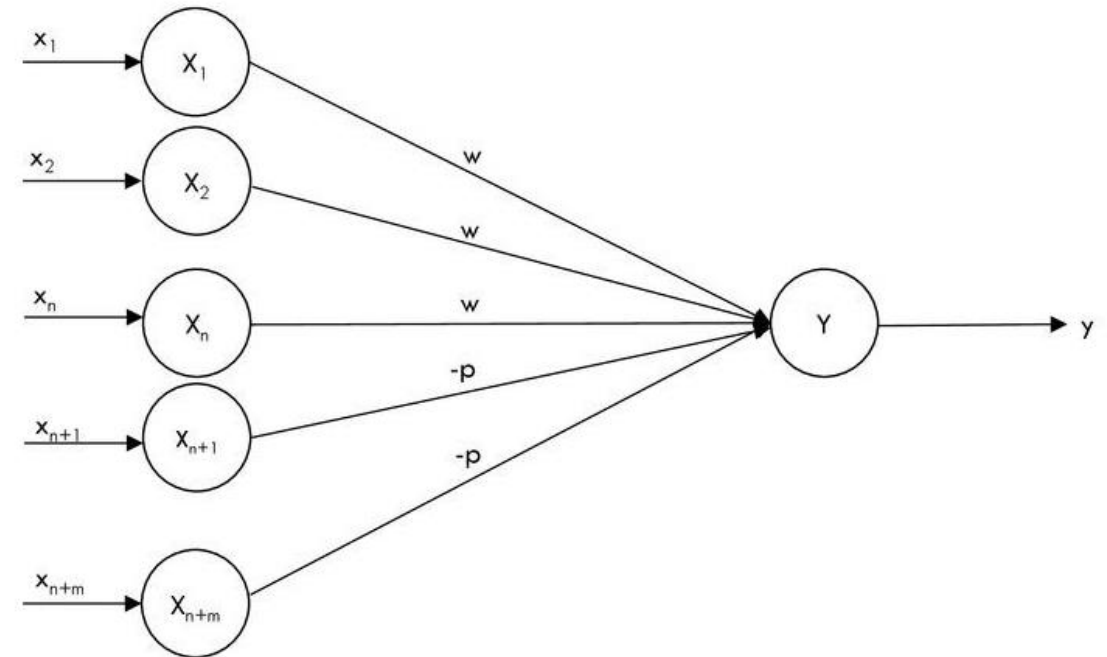
Early Foundations (Pre-1960s)



McCulloch (right) and Pitts (left) in 1949

In 1943, neurophysiologist Warren McCulloch and mathematician Walter Pitts wrote a paper on how neurons might work. In order to describe how neurons in the brain might work, they modeled a simple neural network using electrical circuits.

McCULLOCH-PITTS NEURON: ARCHITECTURE (MP NEURON MODEL)



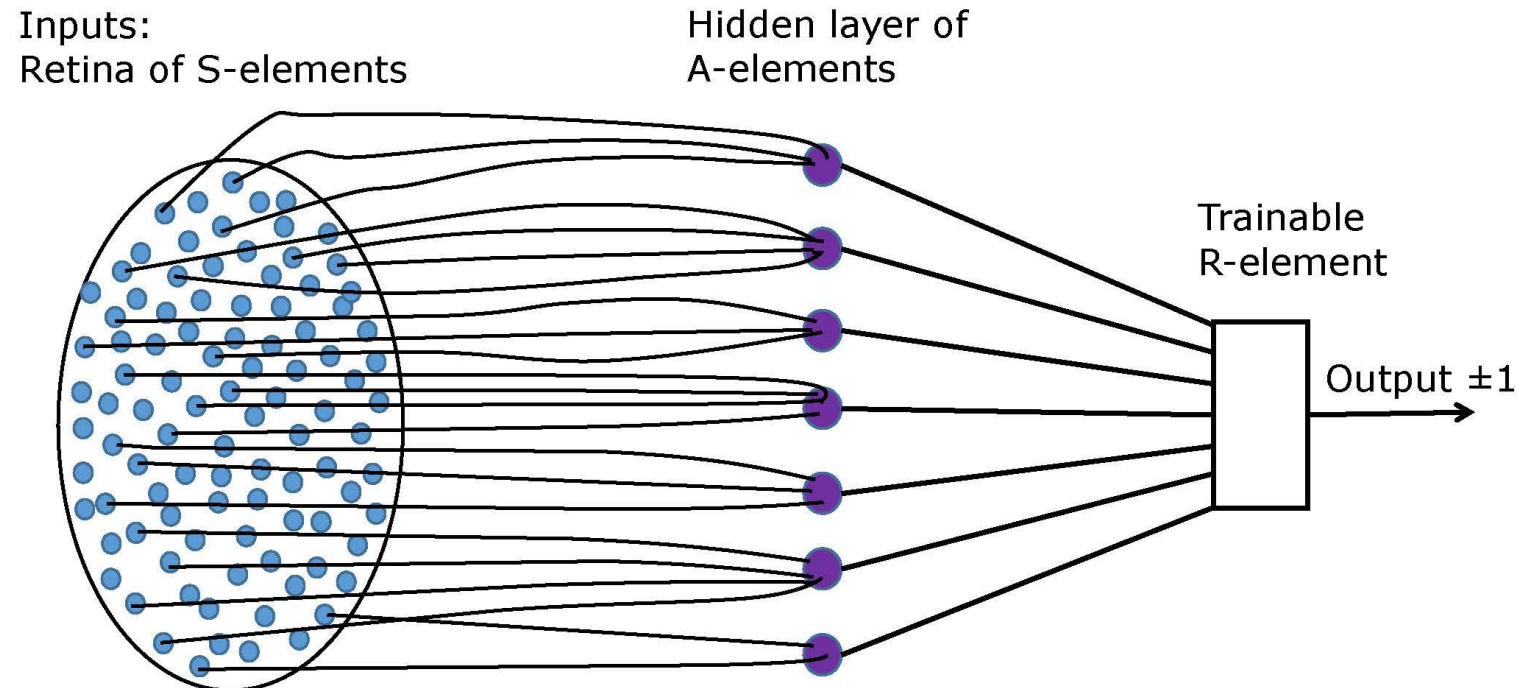
1950s: Early neural networks (McCulloch & Pitts, 1943)

Early Foundations (Pre-1960s)



Rosenblatt in 1950

An elementary Rosenblatt's perceptron. A-units are linear threshold element with fixed input weights. R-unit is also a linear threshold element but with ability to learn according to Rosenblatt's [learning rule](#)

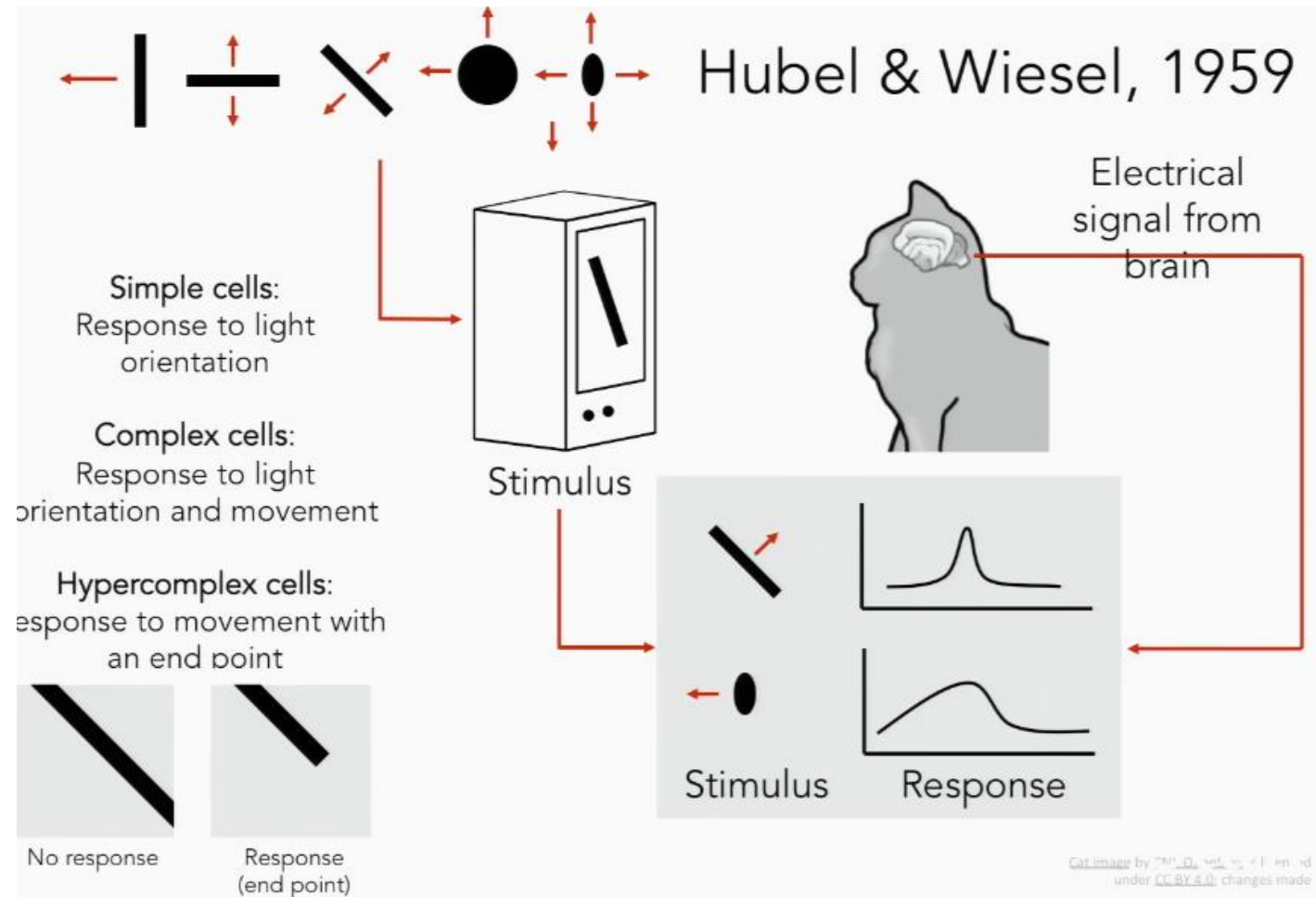


1957: Frank Rosenblatt's Perceptron (early machine learning for vision)

Early Foundations (Pre-1960s)

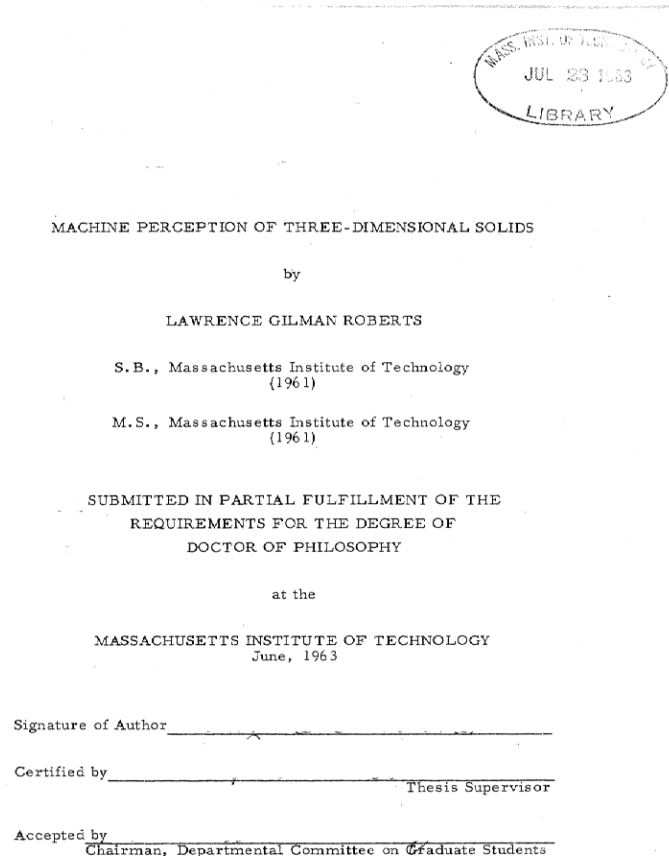


Hubel (left) and Wiesel (right) after receiving news of their Nobel Prize in 1981.



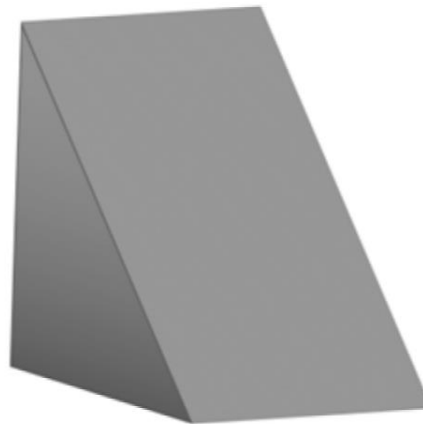
1959: Hubel & Wiesel's cat experiments (biological vision inspiration)

The Birth of Computer Vision (1960s-1970s)

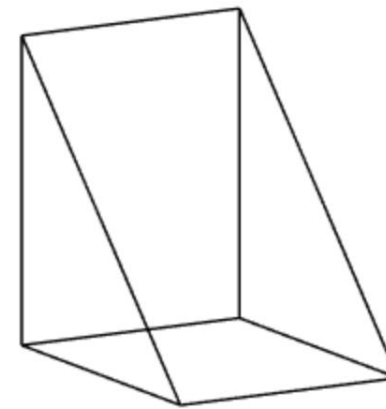


This thesis is often considered foundational because it explored how to extract 3D information from 2D images of blocks (polyhedra).

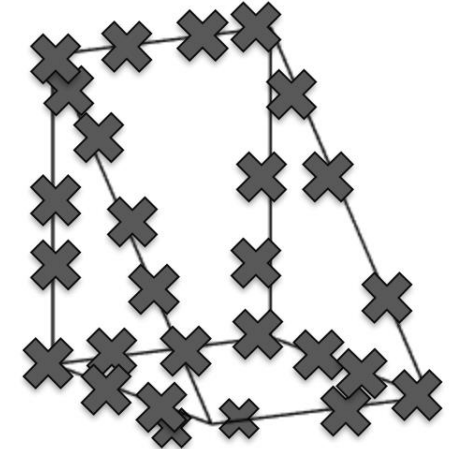
Roberts' work tackled the fundamental challenge of inferring a 3D model from a 2D image, a central theme in computer vision



(a) Original picture



(b) Differentiated picture



(c) Feature points selected

1963: Larry Roberts ("Machine Perception of 3D Solids") – first PhD thesis on CV

The Birth of Computer Vision (1960s-1970s)

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

THE SUMMER VISION PROJECT

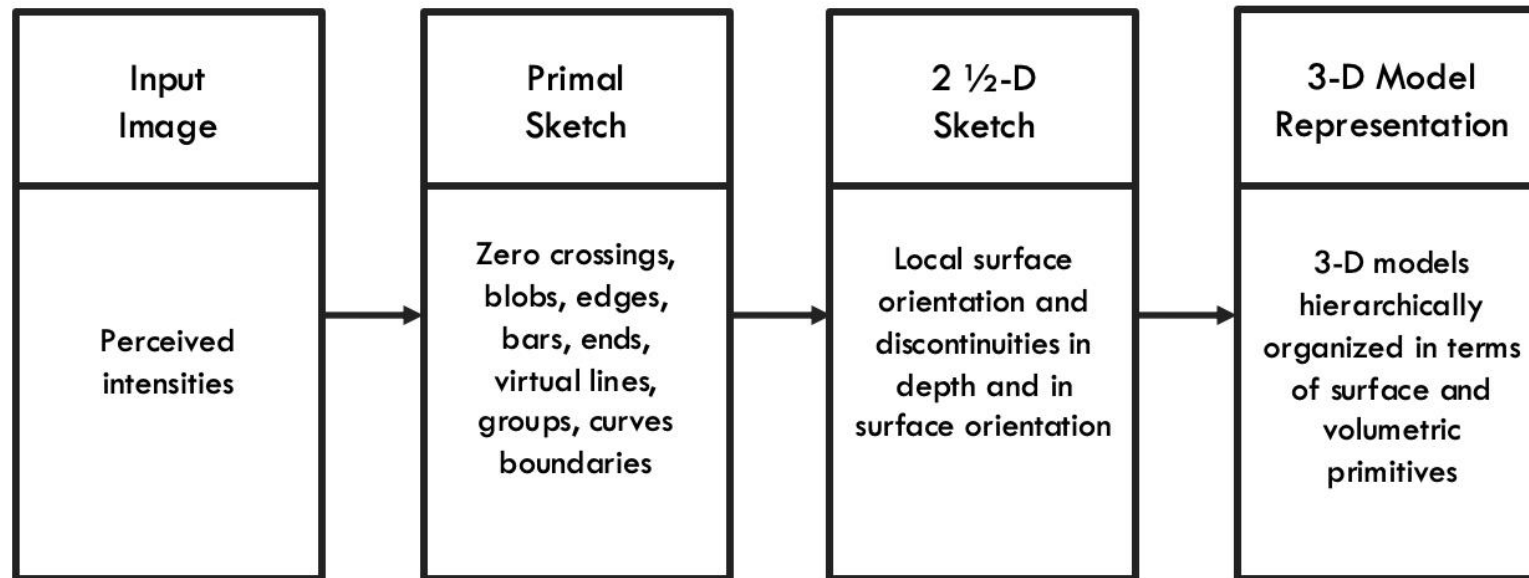
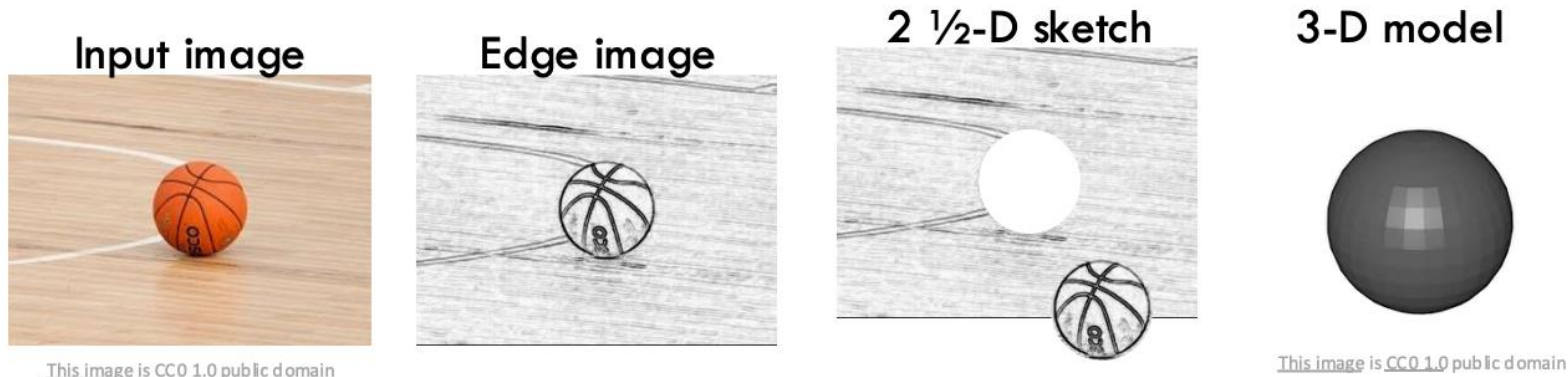
Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition". The basic structure is fixed for the first phase of work extending to some point in July. Everyone is invited to contribute to the discussion of the second phase. Sussman is coordinator of "Vision Project" meetings and should be consulted by anyone who wishes to participate. The primary goal of the project is to construct a system of programs which will divide a vidisector picture into regions such as likely objects, likely background areas and chaos. We shall call this part of its operation FIGURE-GROUND analysis. It will be impossible to do this without considerable analysis of shape and surface properties, so FIGURE-GROUND analysis is really inseparable in practice from the second goal which is REGION DESCRIPTION. The final goal is OBJECT IDENTIFICATION which will actually name objects by matching them with a vocabulary of known objects.

1966: MIT's "Summer Vision Project" – first attempt at object recognition

The Birth of Computer Vision (1960s-1970s)

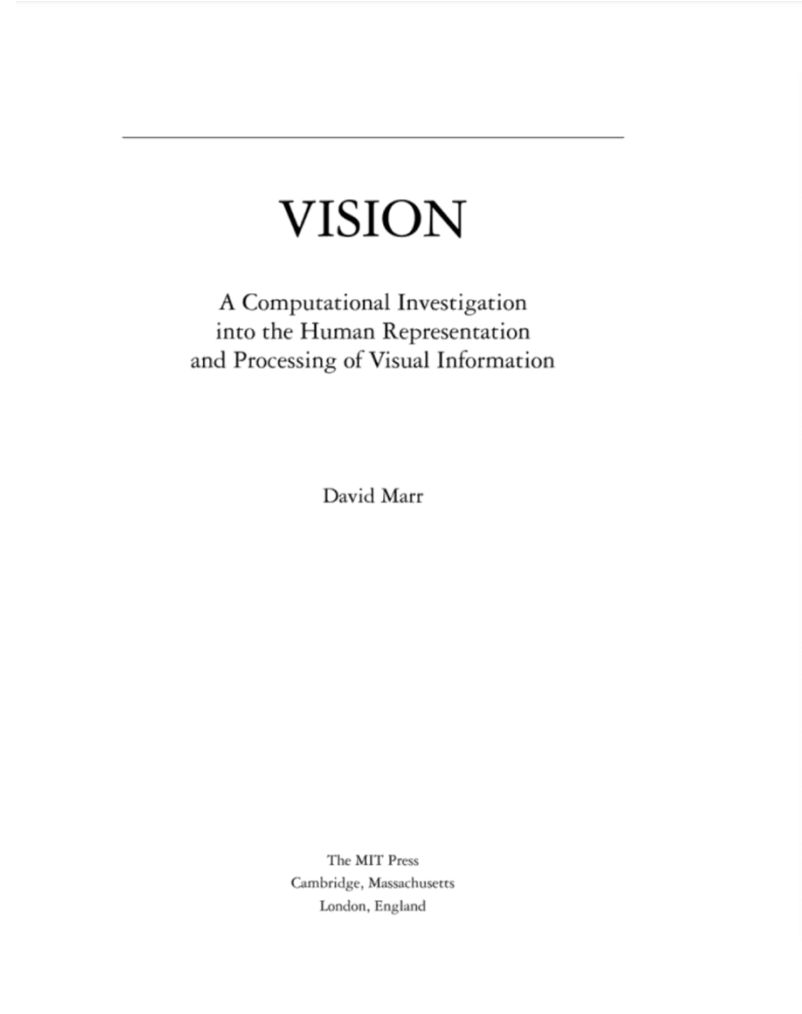
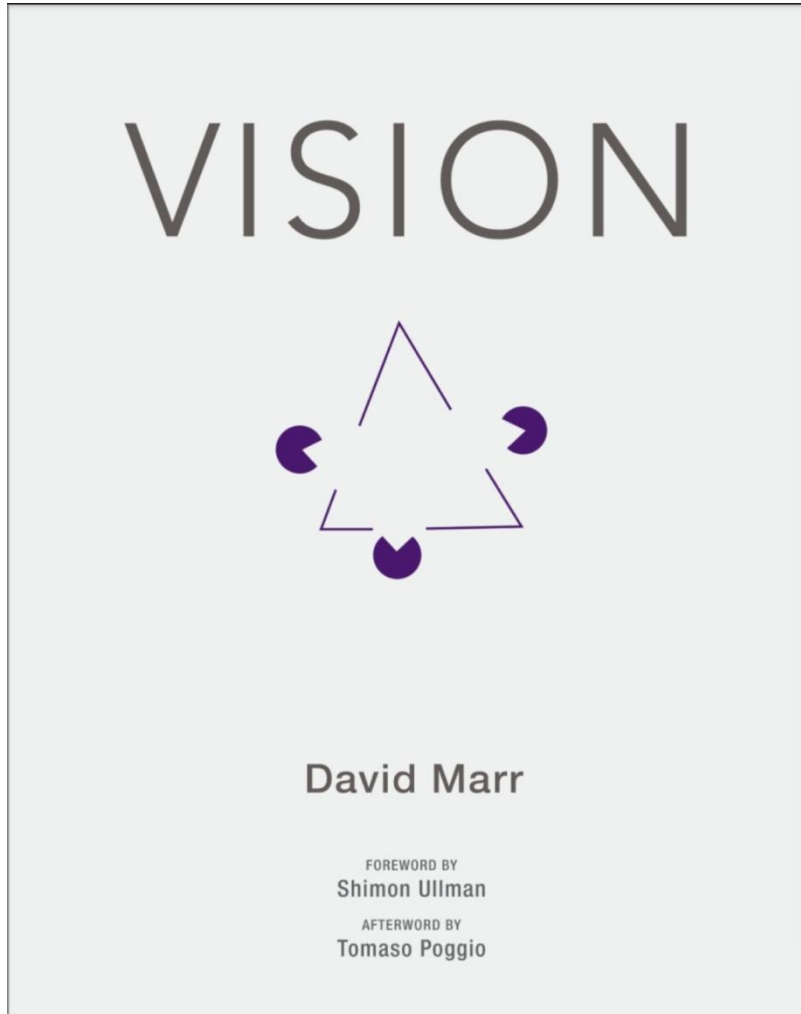


1970s: David Marr's theories (Marr's Paradigm: Primal Sketch → 2.5D → 3D)



The ICCV best-paper award is the Marr Prize,
named after British neuroscientist [David Marr](#)

Geometric & Mathematical Advances (1980s)



Geometric & Mathematical Advances (1980s)

Recognition via Edge Detection



Machine Learning & Statistical Methods (1990s-2010s)

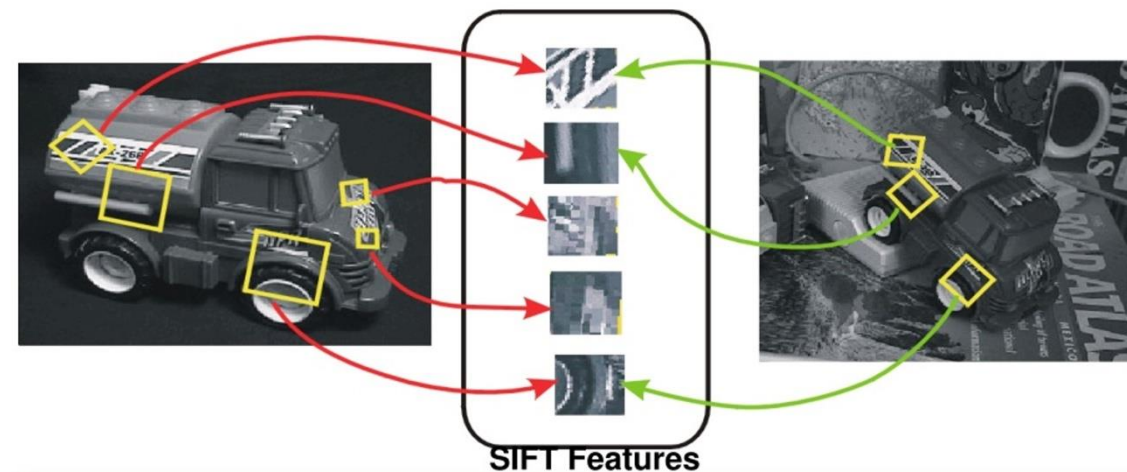


An **eigenface** (*/ˈaɪɡən-/ EYE-gən-*) is the name given to a set of [eigenvectors](#) when used in the [computer vision](#) problem of human [face recognition](#).^[1] The approach of using eigenfaces for [recognition](#) was developed by Sirovich and Kirby and used by [Matthew Turk](#) and [Alex Pentland](#) in face classification.^{[2][3]} The eigenvectors are derived from the [covariance matrix](#) of the [probability distribution](#) over the high-[dimensional vector space](#) of face images. **The eigenfaces themselves form a basis set of all images used to construct the covariance matrix.** This produces dimension reduction by allowing the smaller set of basis images to represent the original training images. **Classification can be achieved by comparing how faces are represented by the basis set.**

Machine Learning & Statistical Methods (1990s-2010s)

Idea of SIFT

- Image content is transformed into local feature coordinates that are invariant to translation, rotation, scale, and other imaging parameters



Recognition via matching

1999: SIFT (Scale-Invariant Feature Transform, David Lowe)

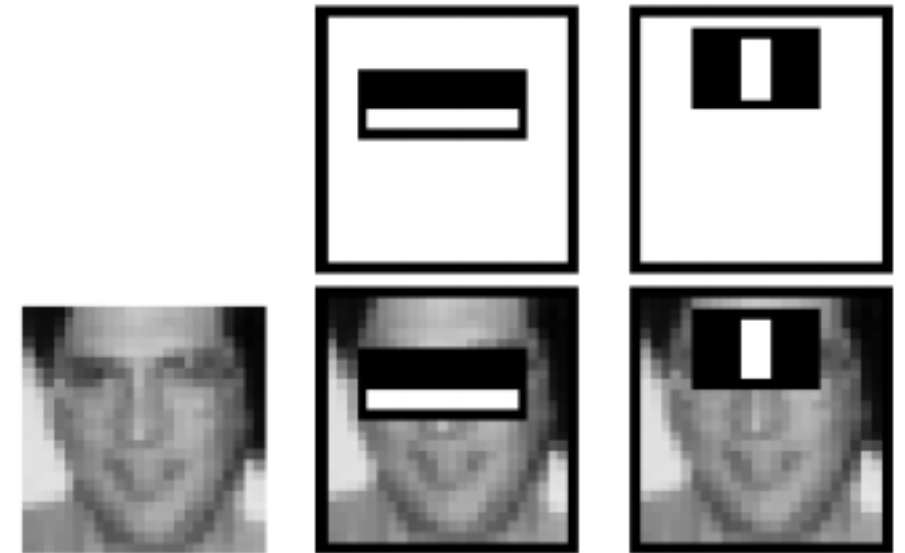
Machine Learning & Statistical Methods (1990s-2010s)

The Viola/Jones Face Detector

- A “paradigmatic” method for real-time object detection
- Training is slow, but detection is very fast
- Key ideas
 - *Integral images* for fast feature evaluation
 - *Boosting* for feature selection
 - *Attentional cascade* for fast rejection of non-face windows



P. Viola and M. Jones. *Rapid object detection using a boosted cascade of simple features*. CVPR 2001.



$$f(x, y) = \sum_i p_b(i) - \sum_i p_w(i)$$

2001: First real-time face detection (Paul Viola & Michael Jones)

Machine Learning & Statistical Methods (1990s-2010s)

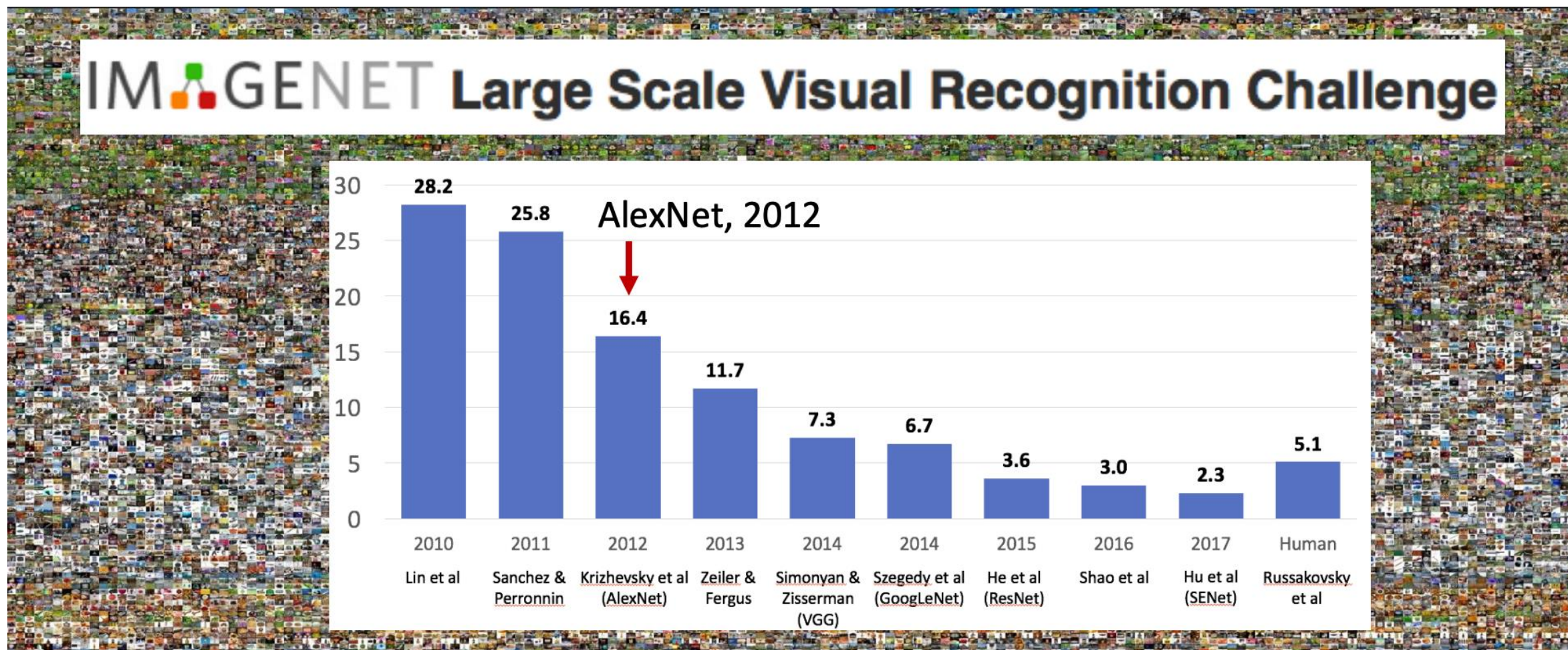


Finding enough labeled data has always been a challenge for Deep Learning community. In 2009 Fei-Fei Li, a professor at Stanford, launches ImageNet which is a database of 14 million labeled images. It would serve as a benchmark for the deep learning researchers who would participate in ImageNet competitions (ILSVRC) every year.



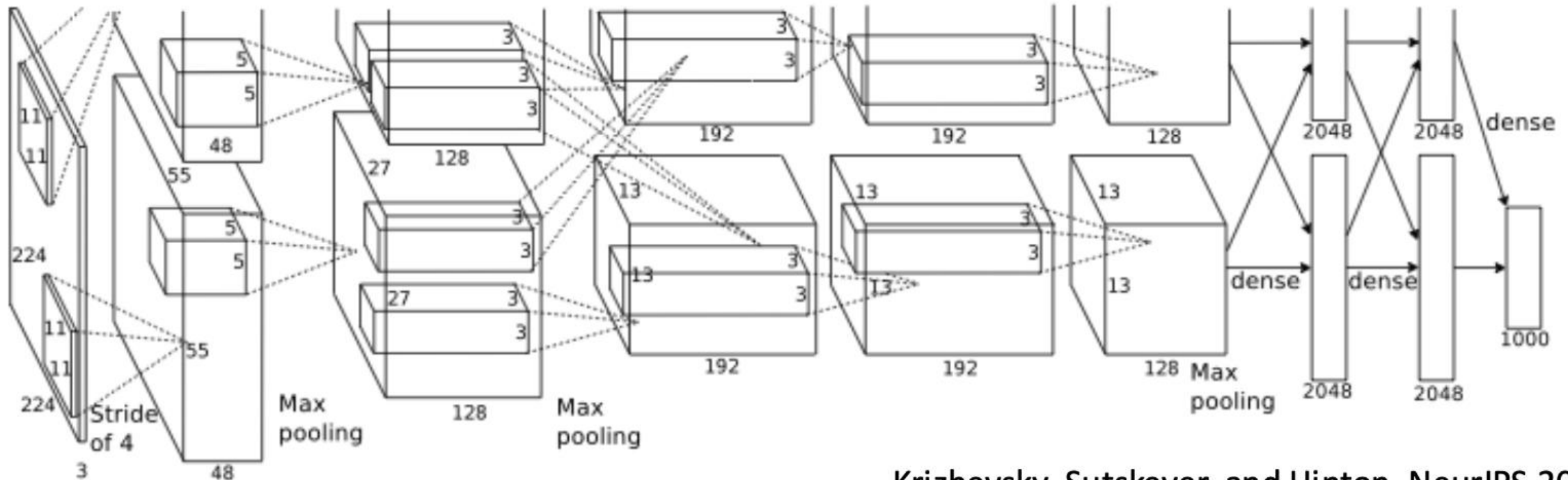
2009: Introduction of ImageNet (Fei-Fei Li et al.)

Deep Learning Revolution (2010s-Present)



2012: AlexNet (Krizhevsky et al.) wins ImageNet, starts CNN dominance

Deep Learning Revolution (2010s-Present)



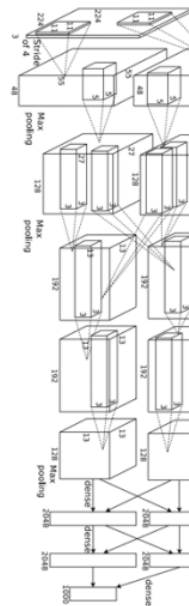
Krizhevsky, Sutskever, and Hinton, NeurIPS 2012

2012: AlexNet (Krizhevsky et al.) wins ImageNet, starts CNN dominance

Deep Learning Revolution (2010s-Present)

Year 2012

SuperVision



[Krizhevsky NIPS 2012]

Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

Year 2014

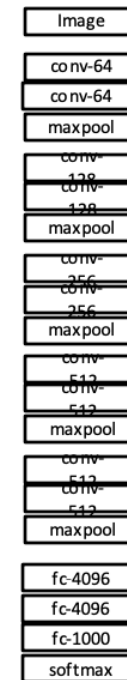
GoogLeNet

- Pooling
- Convolution
- Softmax
- Other



[Szegedy arxiv 2014]

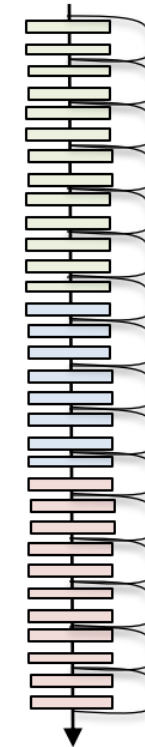
VGG



[Simonyan arxiv 2014]

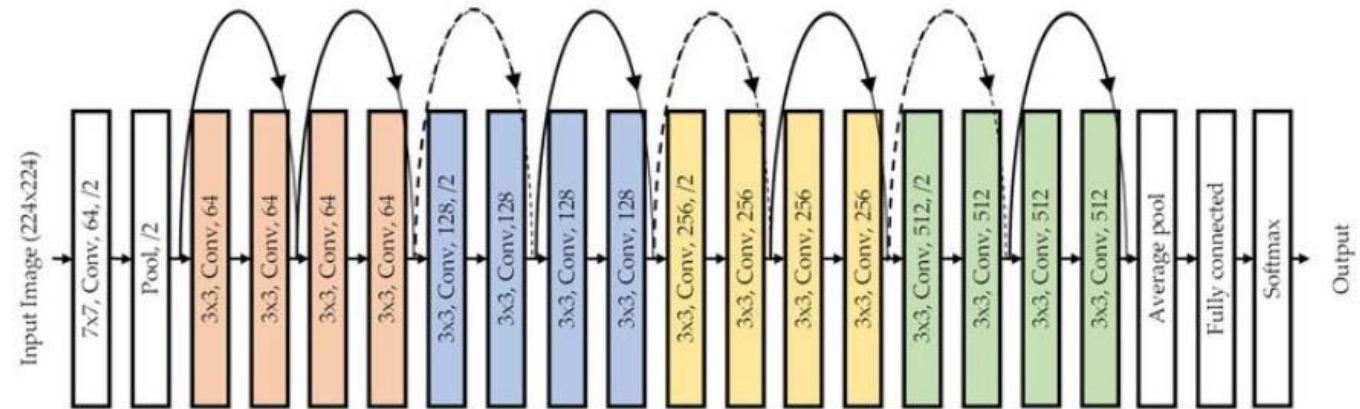
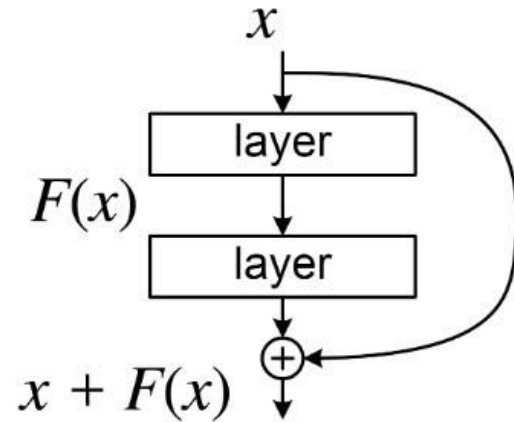
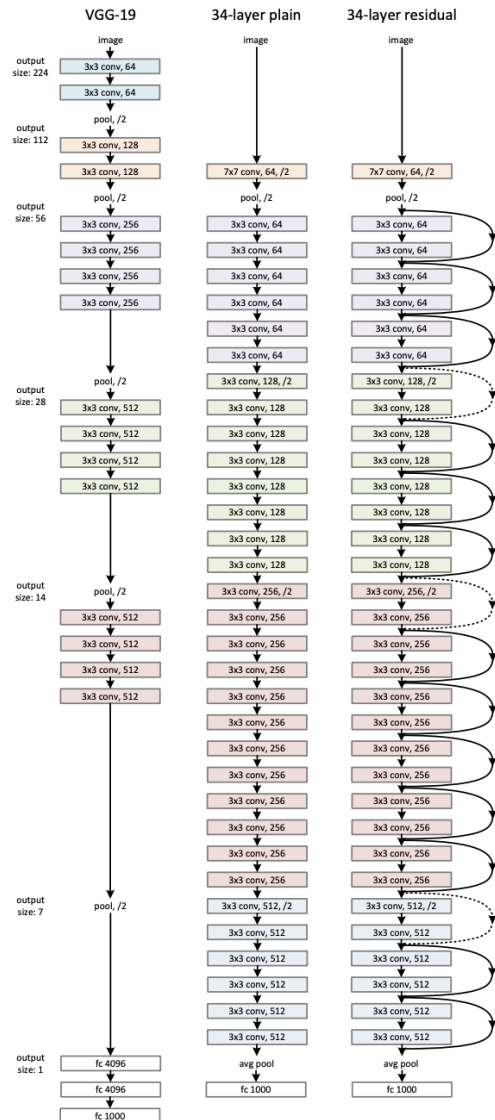
Year 2015

MSRA



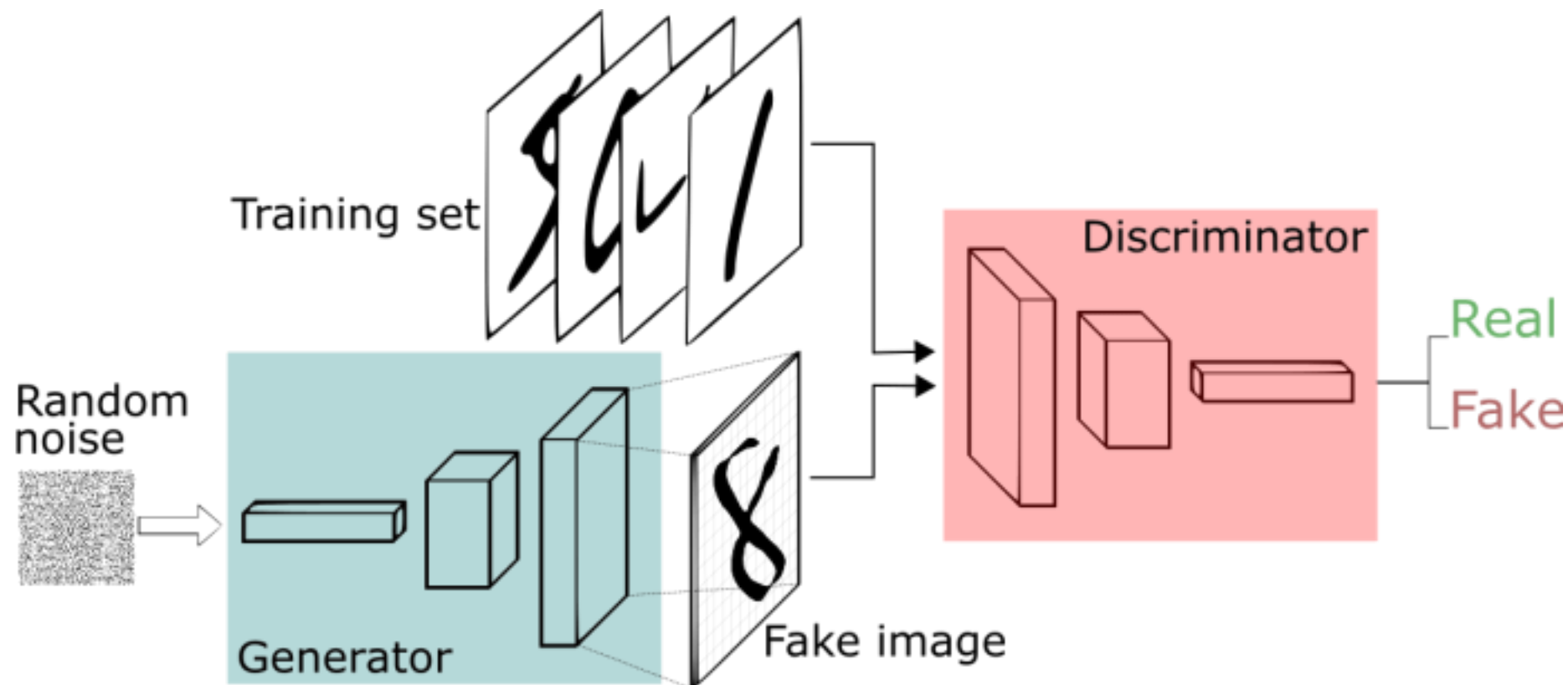
[He ICCV 2015]

Deep Learning Revolution (2010s-Present)



2015: ResNet (He et al.) enables very deep networks

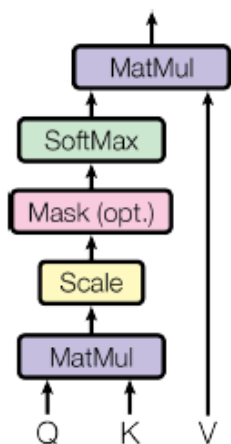
Deep Learning Revolution (2010s-Present)



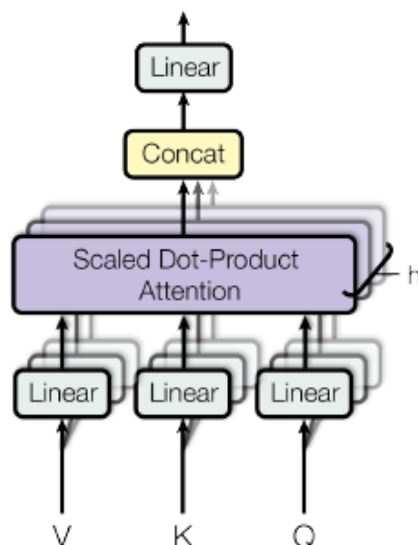
2014: Generative Adversarial Networks (GANs, Ian Goodfellow)

Deep Learning Revolution (2010s-Present)

Scaled Dot-Product Attention

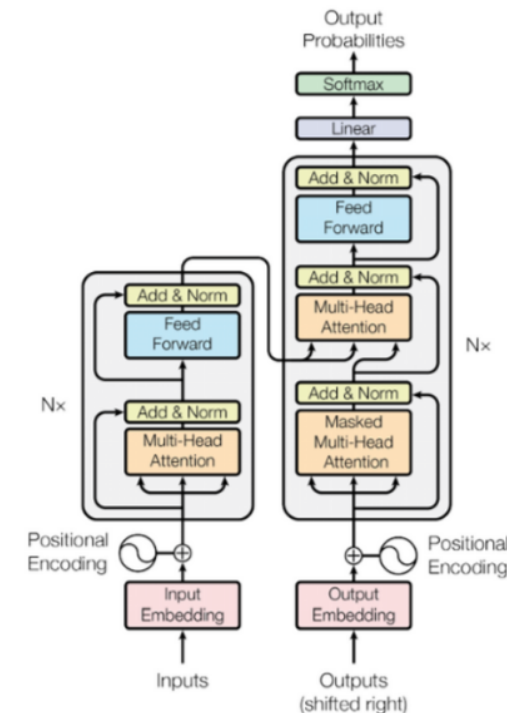


Multi-Head Attention

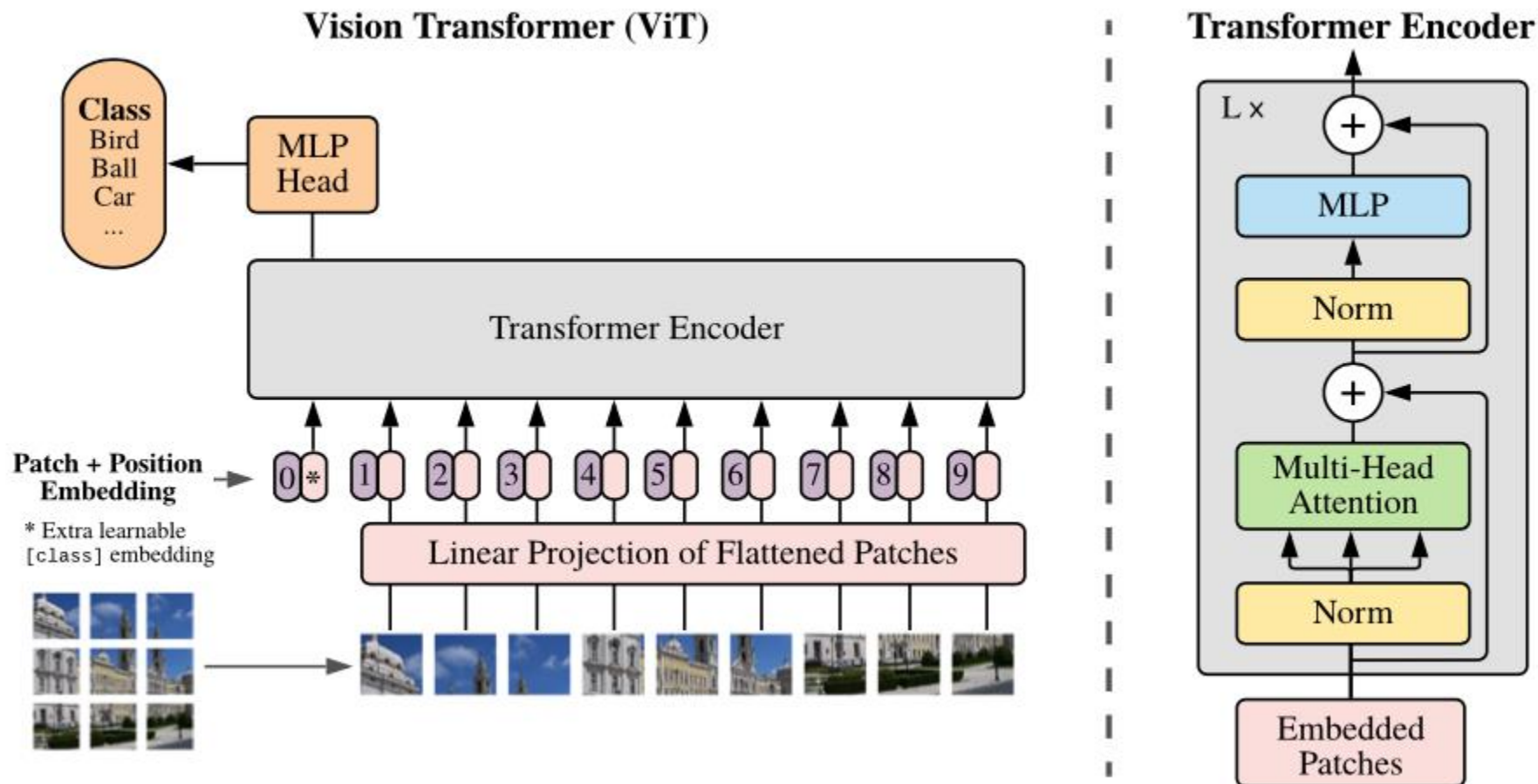


Transformer

Attention Is All You Need



Deep Learning Revolution (2010s-Present)



Introduction to Computer Vision (CV)

- What is Computer Vision
- Brief History of Computer Vision
- **Key Research Areas of Computer Vision**
- Applications of Computer Vision
- Why Study Computer Vision

Low-level Vision (Pixel-based Processing)

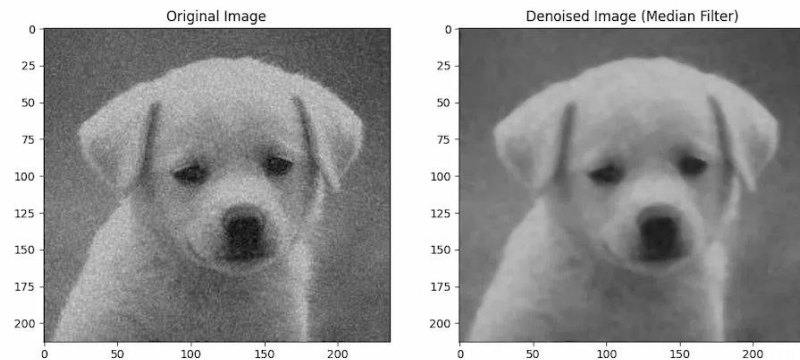
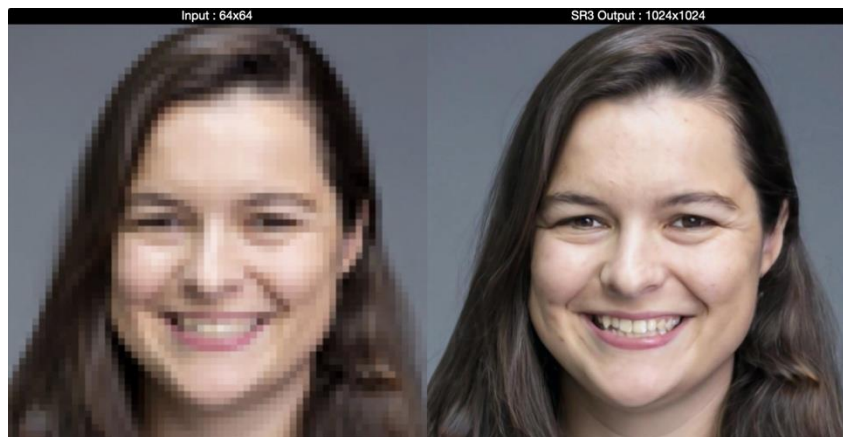


Image denoising (removing noise)

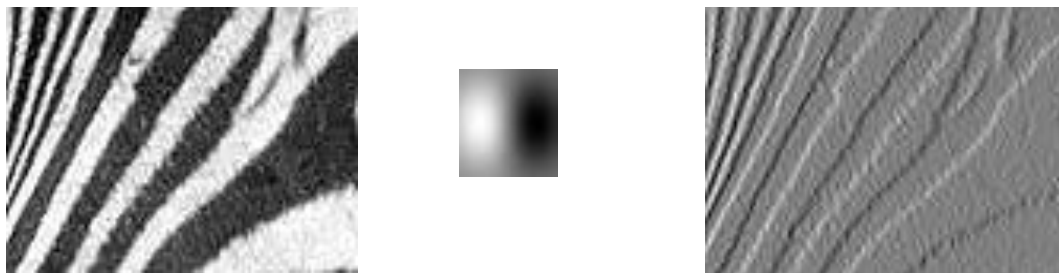


Super-resolution (enhancing low-res images)

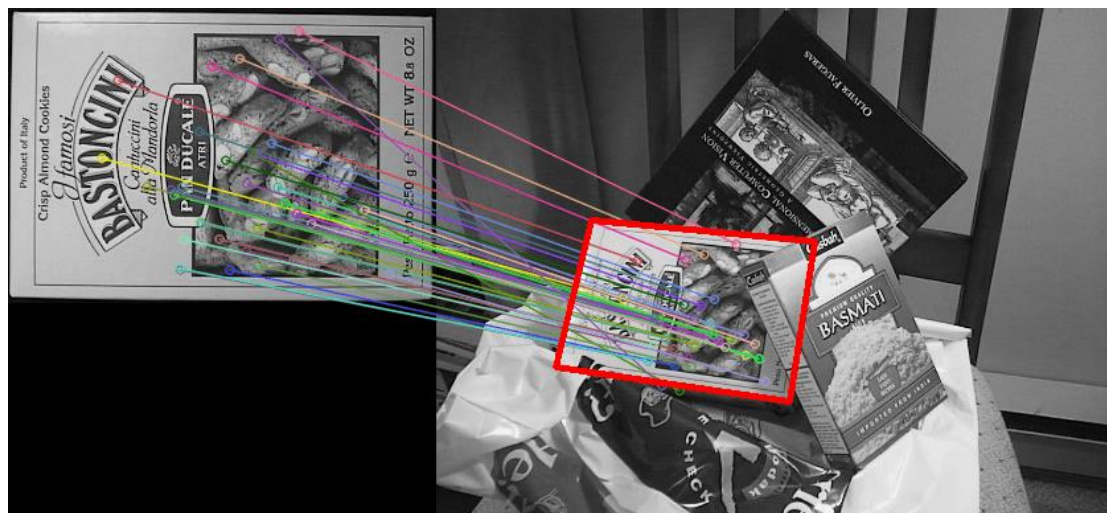


Colorization (adding color to grayscale images)

Mid-level Vision (Feature Extraction & Representation)



Filtering, edge detection



Feature matching (SIFT, ORB, deep descriptors)



(a) Image



(b) Semantic segmentation



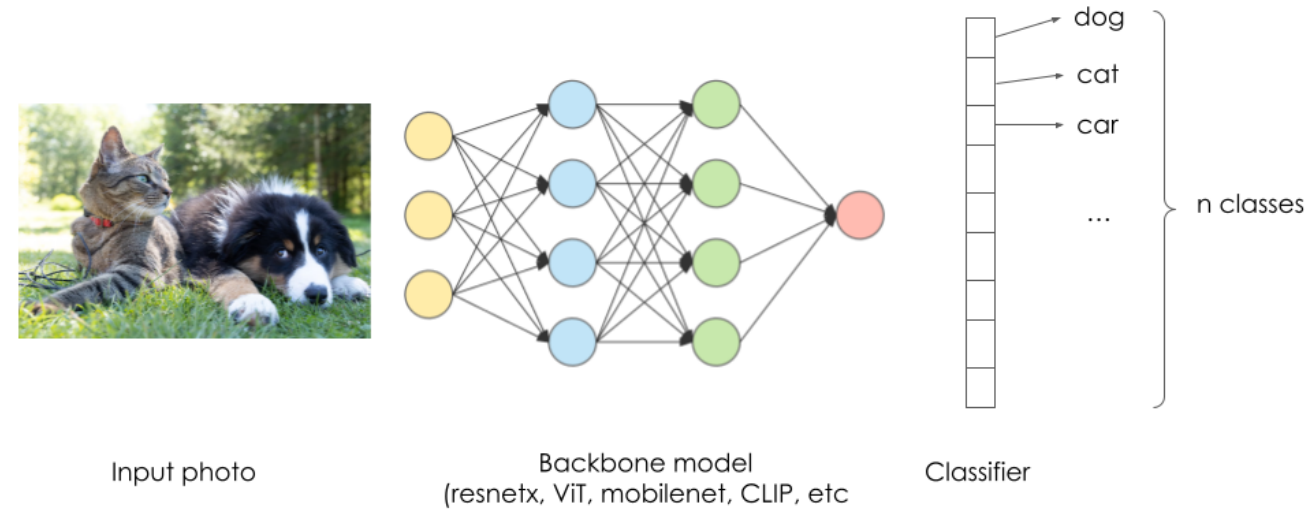
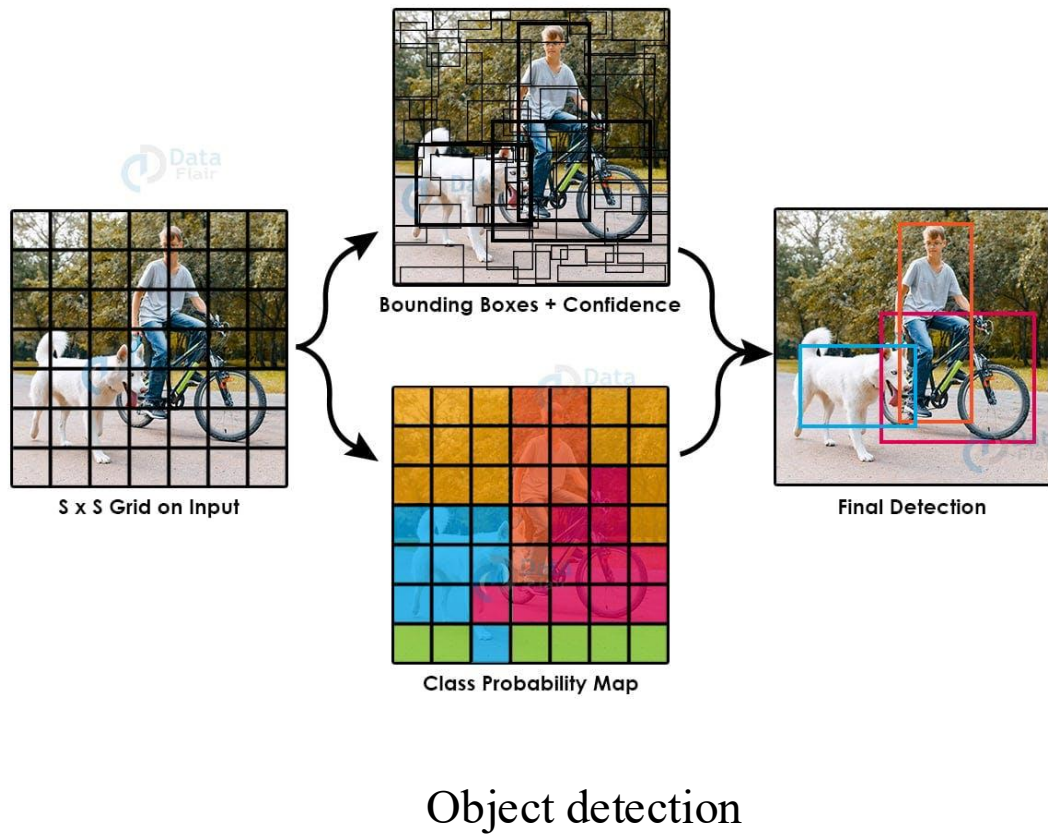
(c) Instance segmentation



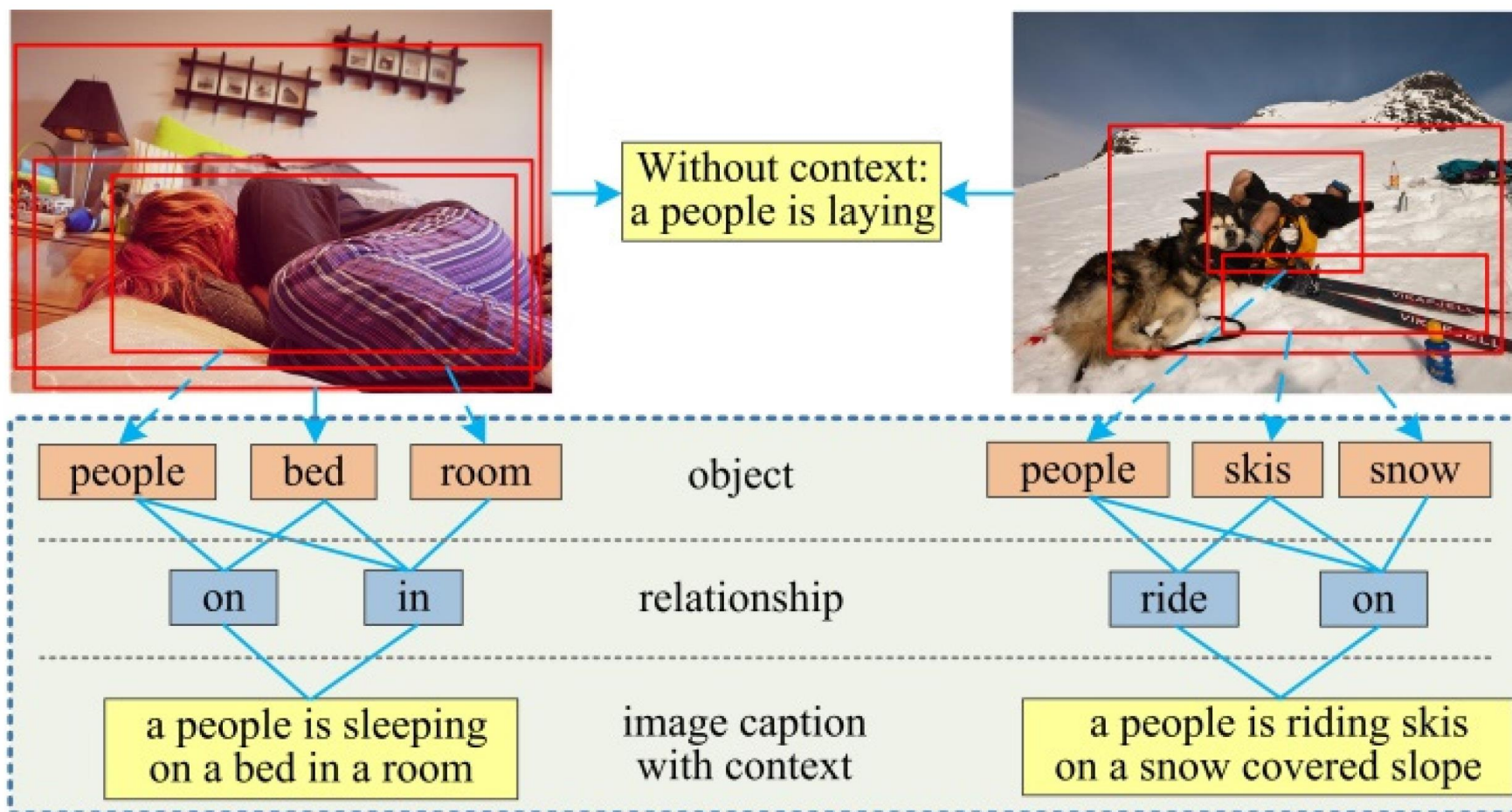
(d) Panoptic segmentation

Image segmentation (semantic/instance segmentation)

High-level Vision (Understanding & Interpretation)

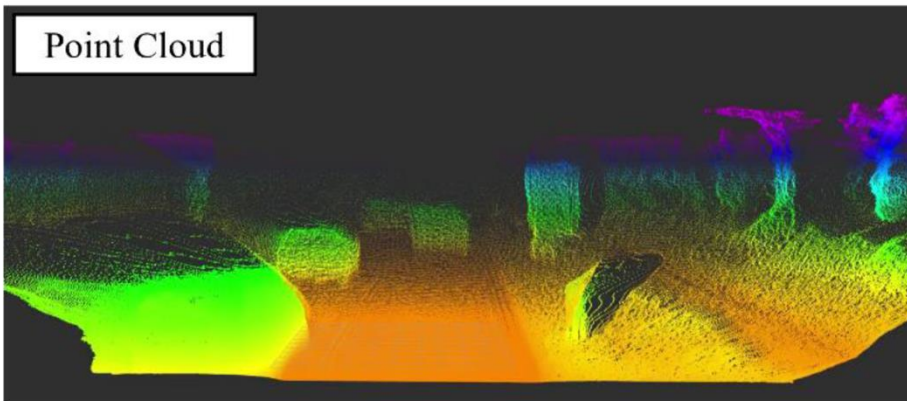
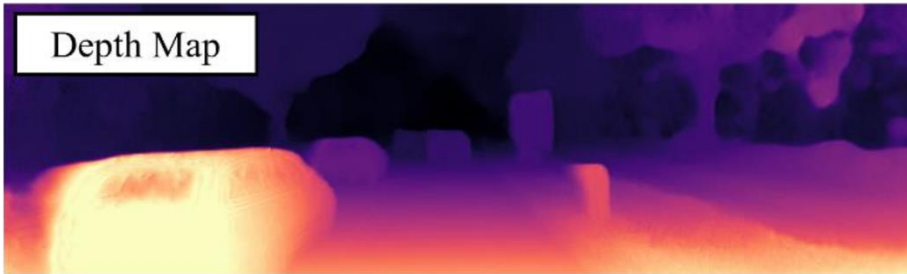


High-level Vision (Understanding & Interpretation)

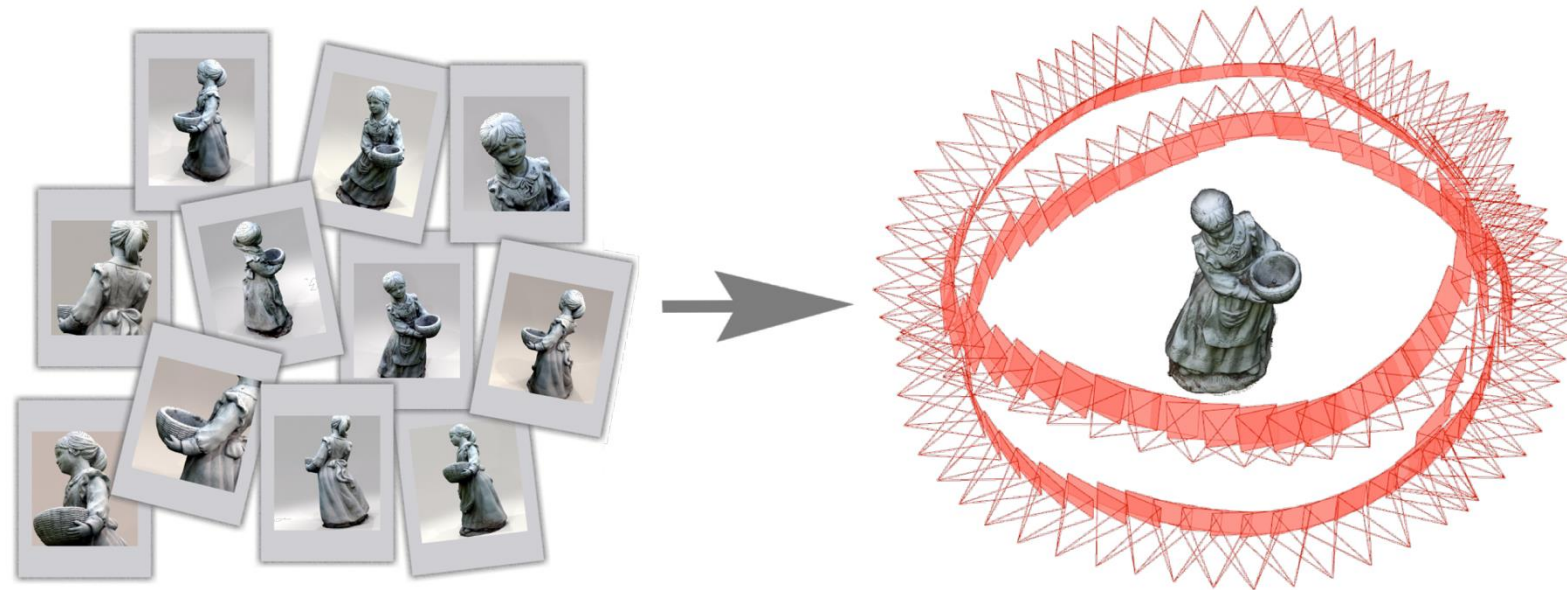


Scene understanding (captioning, relationship detection)

3D Vision & Reconstruction

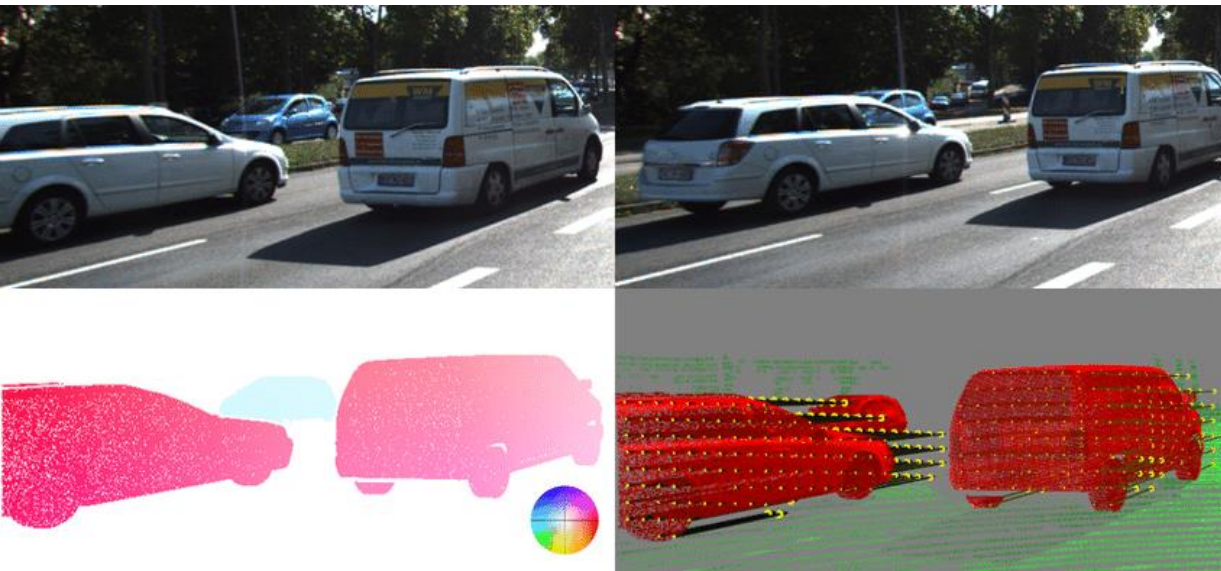


Stereo vision (depth estimation from multiple images)

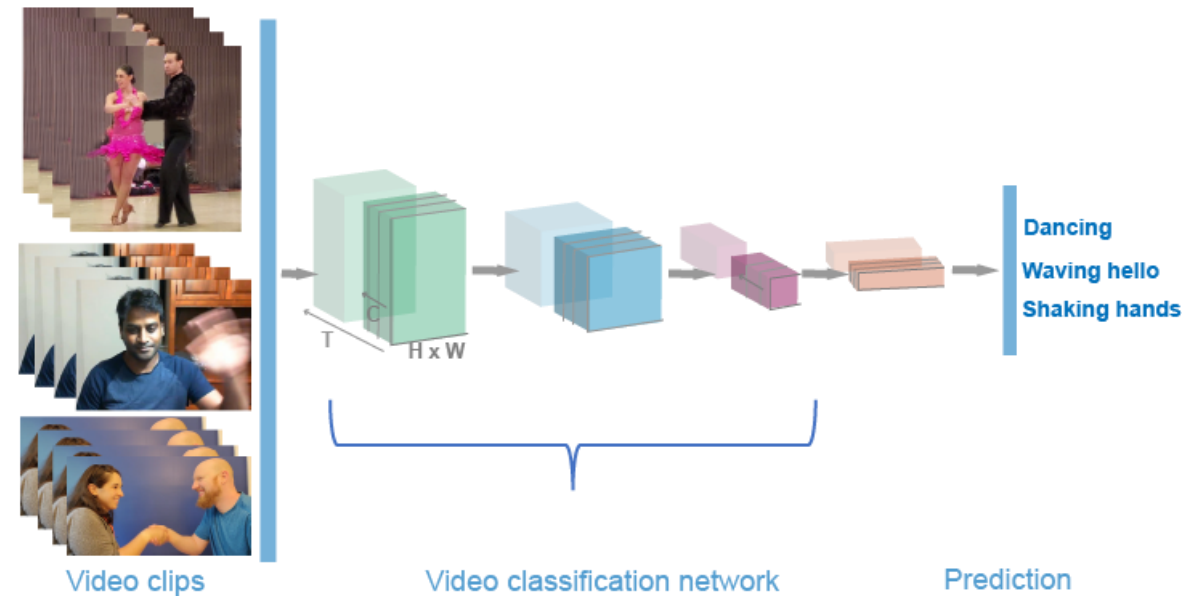


Structure-from-Motion (SfM) (3D reconstruction from multiple images)

Video Analysis & Motion

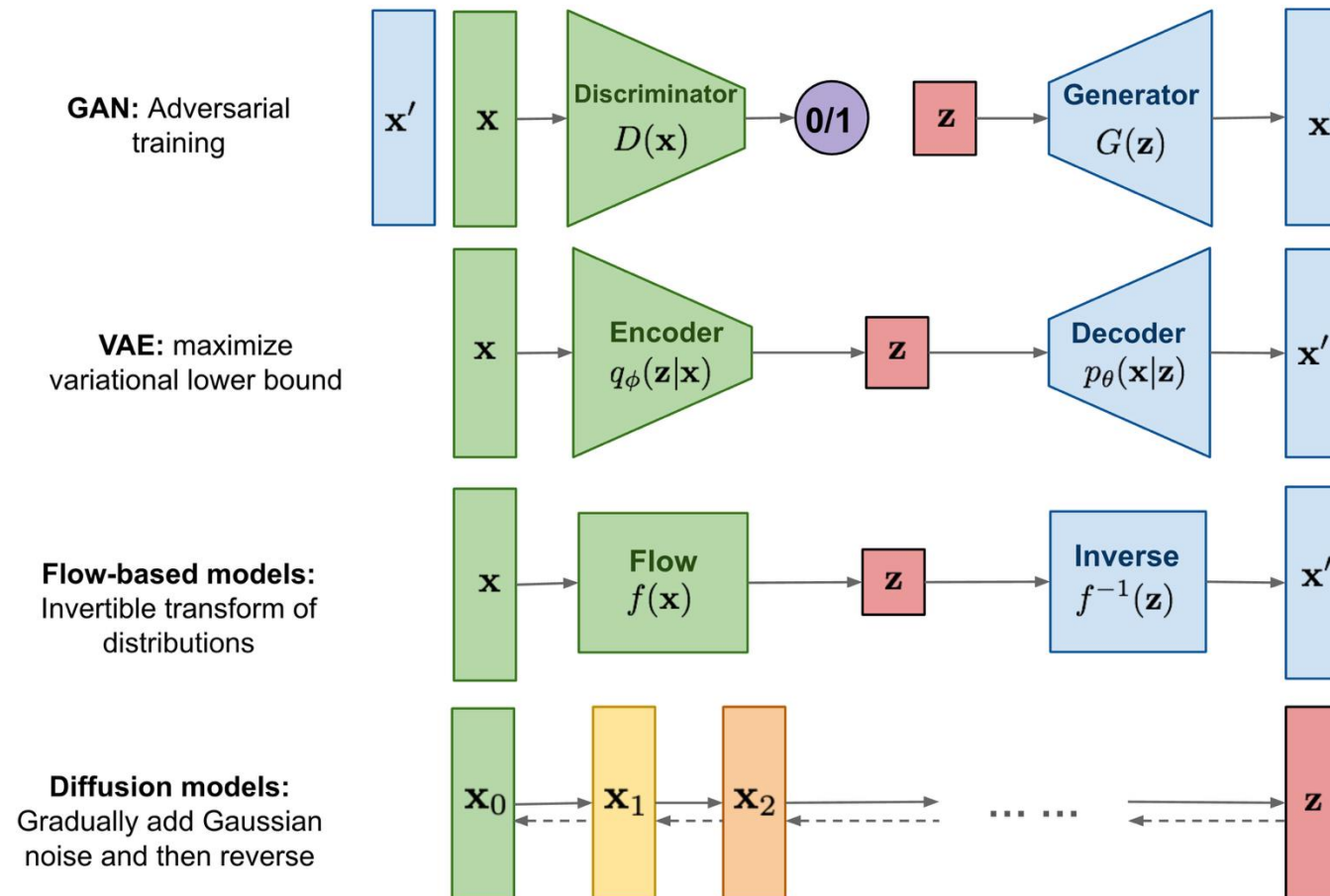


Optical flow (motion estimation between frames)

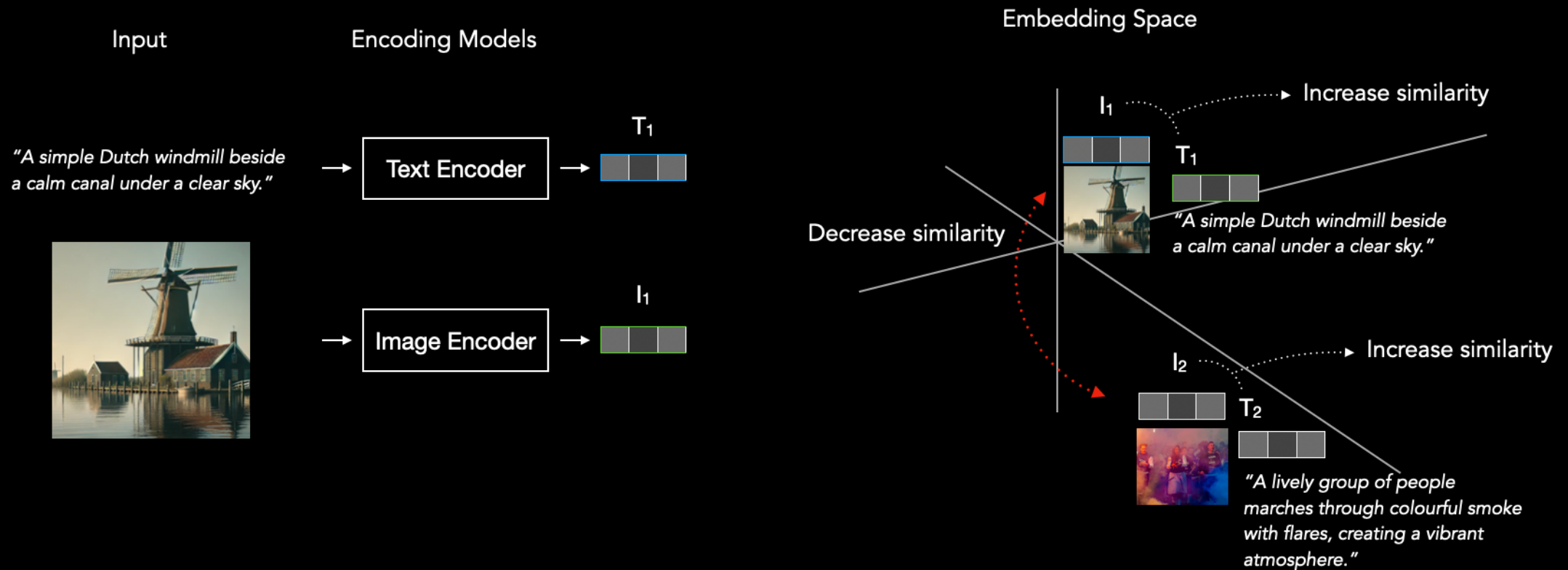


Action recognition (classifying human activities)

Generative Models in CV



Multimodal Learning



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Automated Visual Inspection



Object Recognition



Toshiba Tech IS-910T

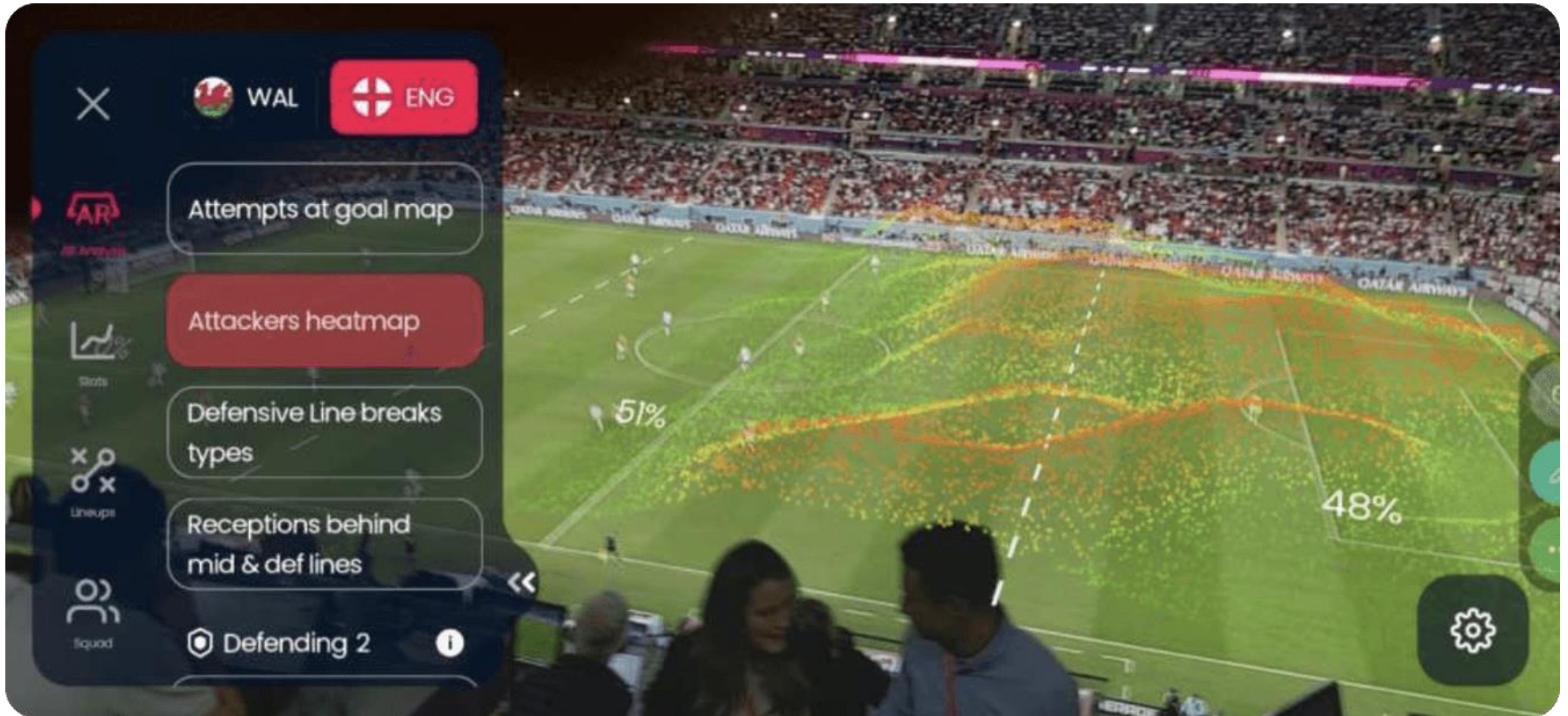
2013



DataLogic LaneHawk LH4000

2012

Computer Vision in Sports



Smart Car

▶▶ manufacturer products consumer products ◀◀

Our Vision. Your Safety.

rear looking camera

forward looking camera

side looking camera

▶ **EyeQ** Vision on a Chip

▶ **Vision Applications**
Road, Vehicle, Pedestrian Protection and more

▶ **AWS** Advance Warning System

▶ **News**

▶ Mobileye Advanced Technologies Power Volvo Cars World First Collision Warning With Auto Brake System

▶ Volvo: New Collision Warning with Auto Brake Helps Prevent Rear-end

▶ all news

▶ **Events**

▶ Mobileye at Equip Auto, Paris, France

▶ Mobileye at SEMA, Las Vegas, NV

▶ read more

- [Mobileye](#)
- Tesla Autopilot
- Safety features in many cars

Smart Car

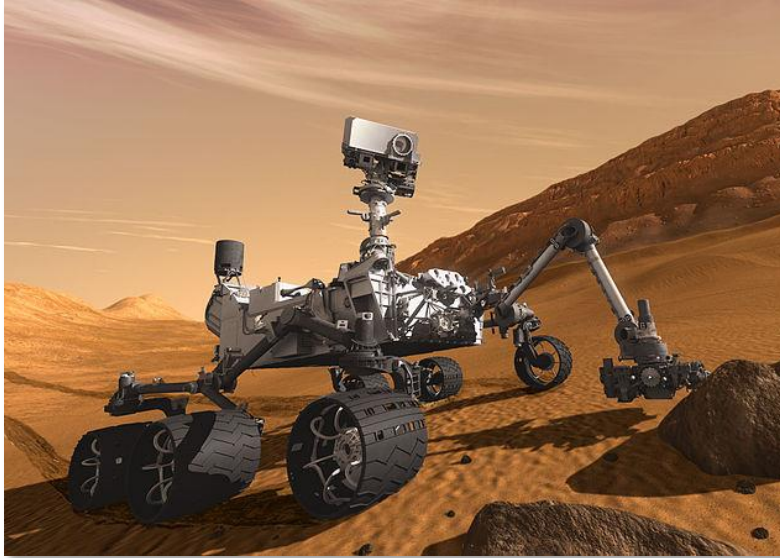


Self-driving Car



Waymo

Robotics



NASA's Mars Curiosity Rover
[https://en.wikipedia.org/wiki/Curiosity_\(rover\)](https://en.wikipedia.org/wiki/Curiosity_(rover))



Amazon Picking Challenge
<http://www.robocup2016.org/en/events/amazon-picking-challenge/>

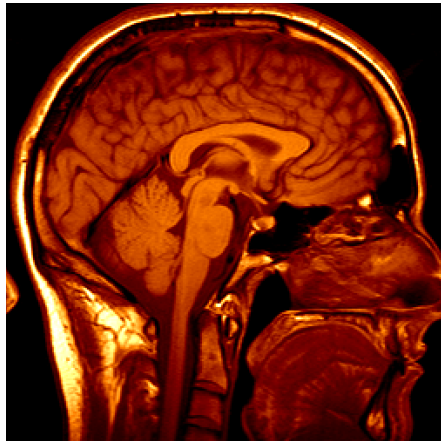


Amazon Prime Air

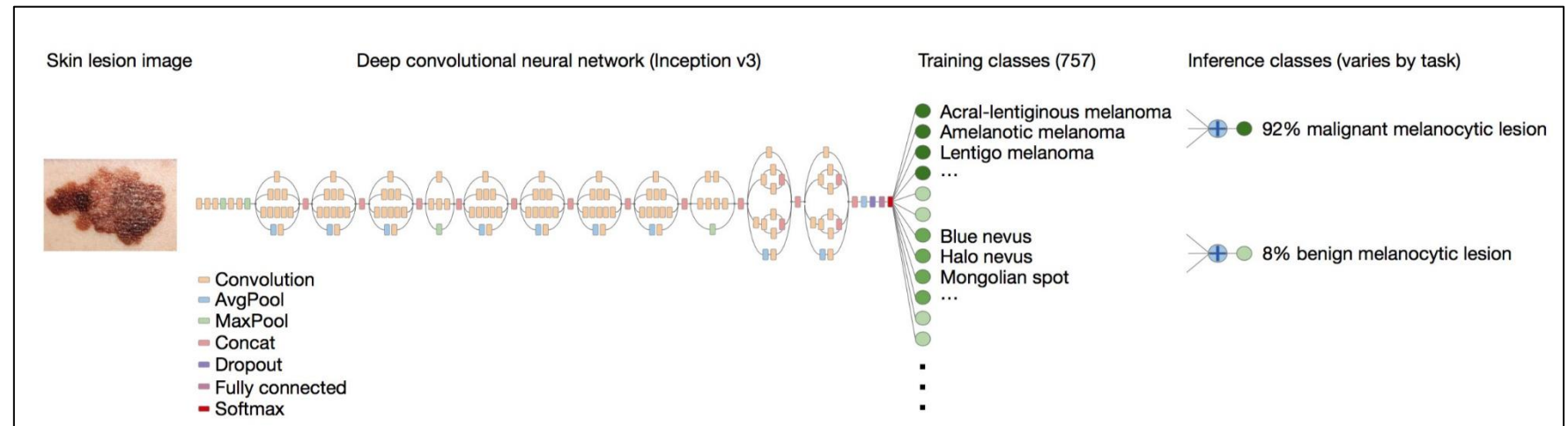


Amazon Scout

Medical Imaging

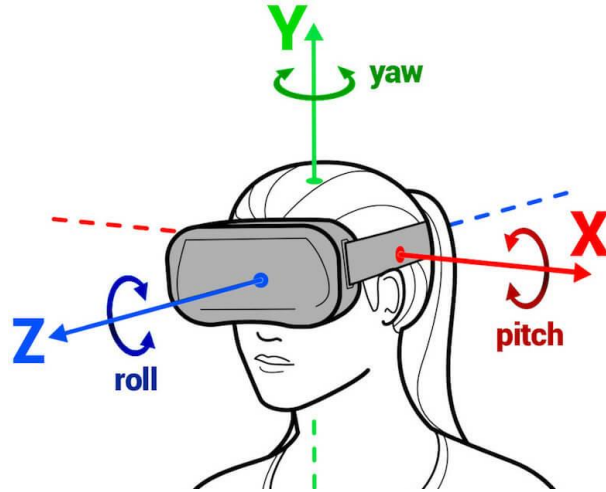


3D imaging
(MRI, CT)



Skin cancer classification with deep learning
<https://cs.stanford.edu/people/esteva/nature/>

Virtual & Augmented Reality



6DoF head tracking



Hand & body tracking



3D scene understanding



3D-360 video capture

INVESTING 3/25/2014 @ 5:43PM | 70,399 views

Facebook Buys Oculus, Virtual Reality Gaming Startup, For \$2 Billion

[+ Comment Now](#) [+ Follow Comments](#)



XR

XR - Extended Reality

XR is an umbrella term that covers VR, AR, and MR



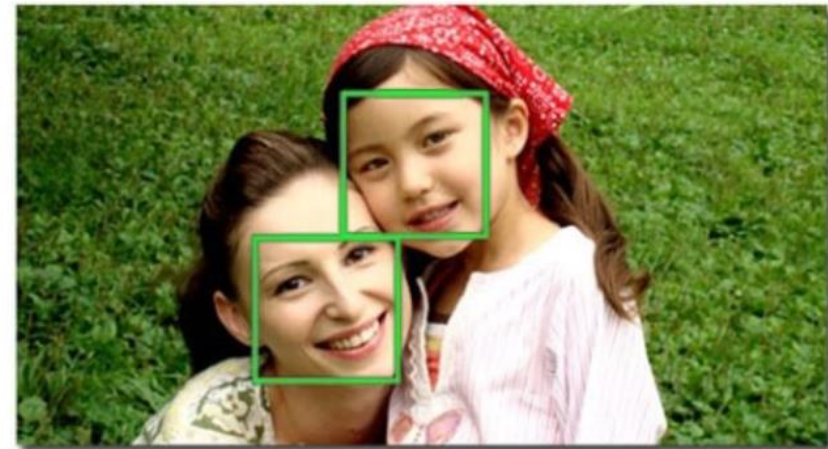
Face detection



Sony Cyber-shot

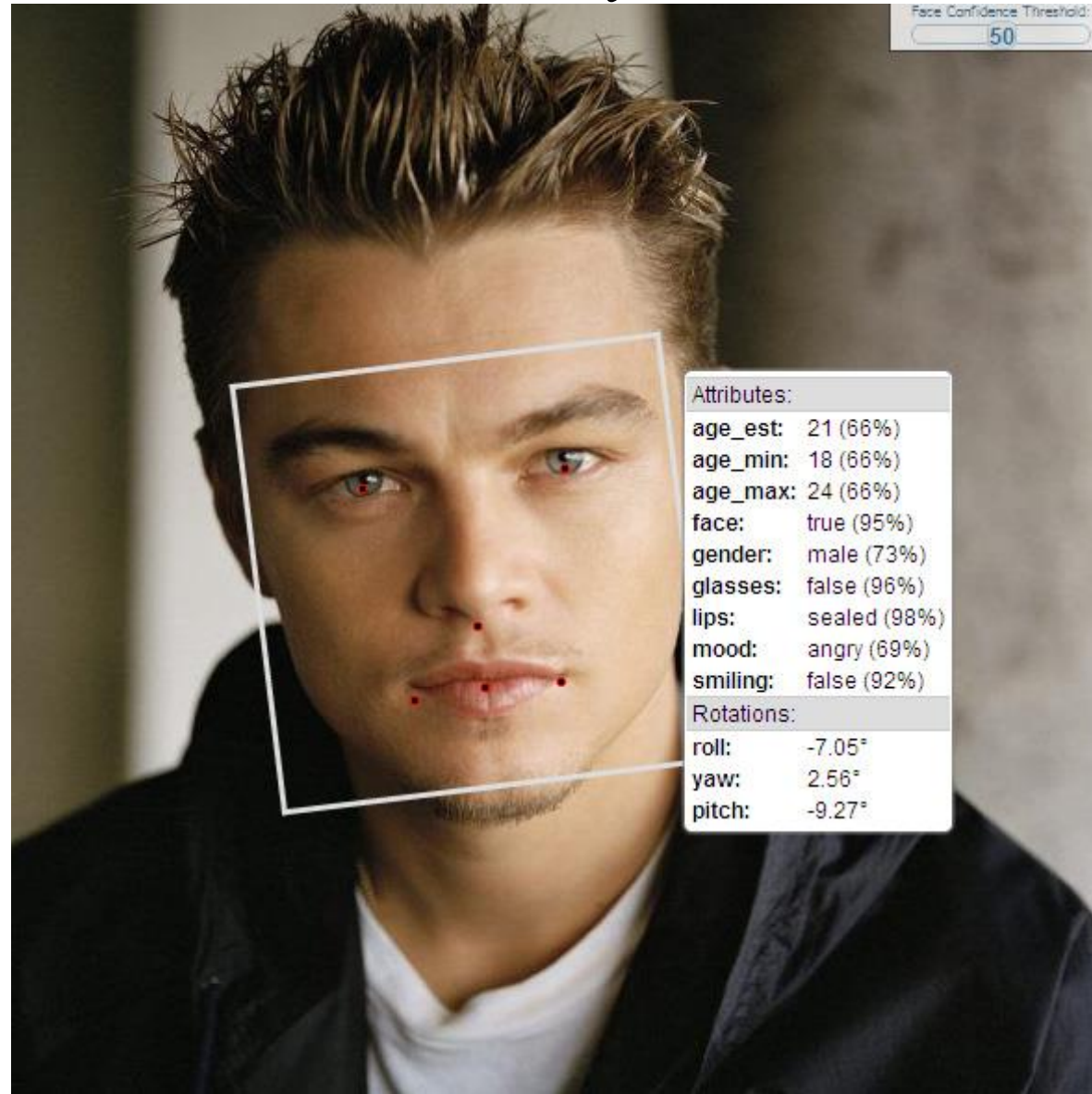


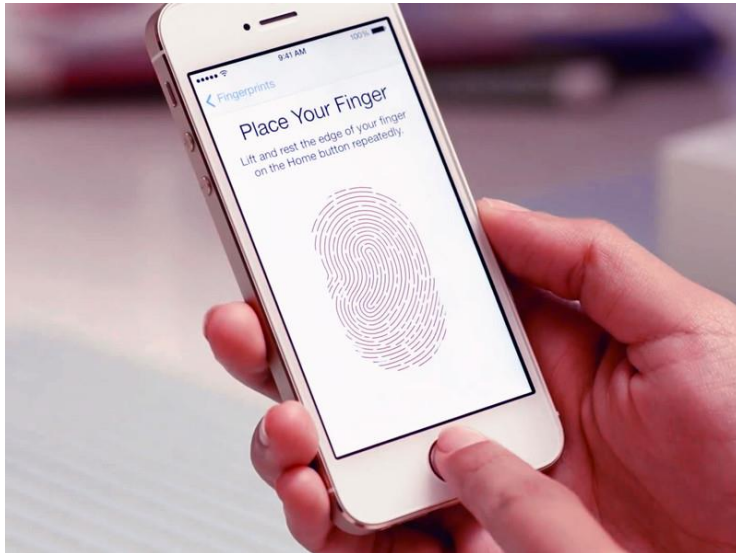
Age recognition



Smile recognition

Face Analysis





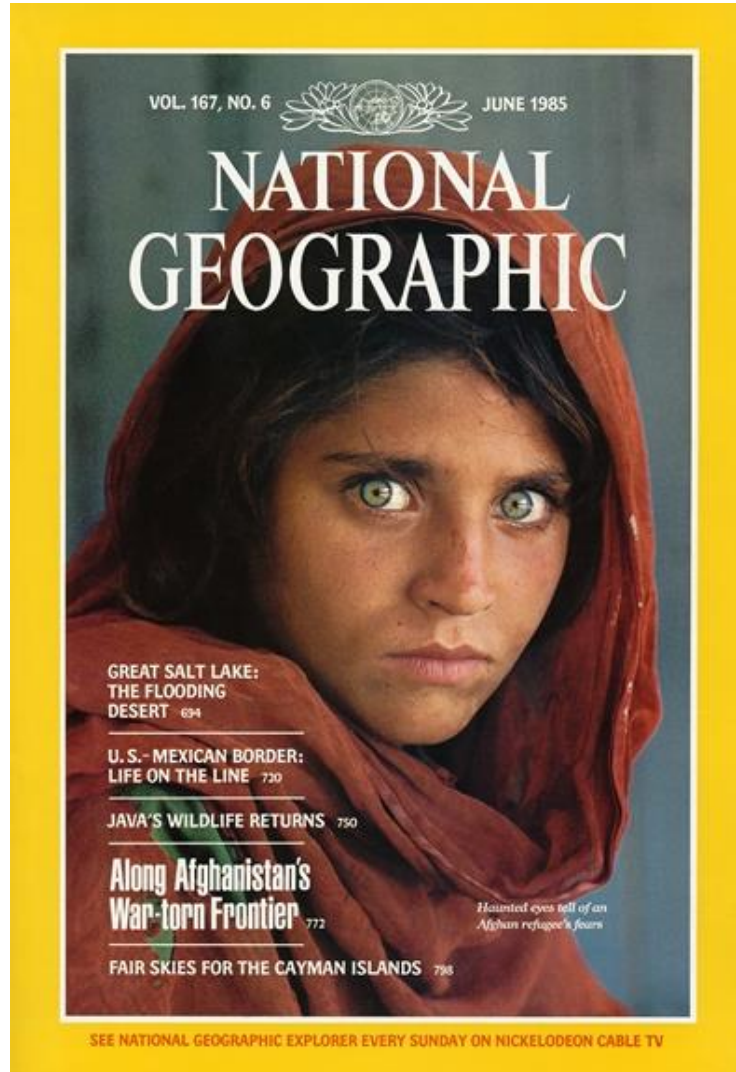
Fingerprint scanners on many new smartphones and other devices



Face unlock on Apple iPhone X
See also <http://www.sensiblevision.com/>

Log in without password

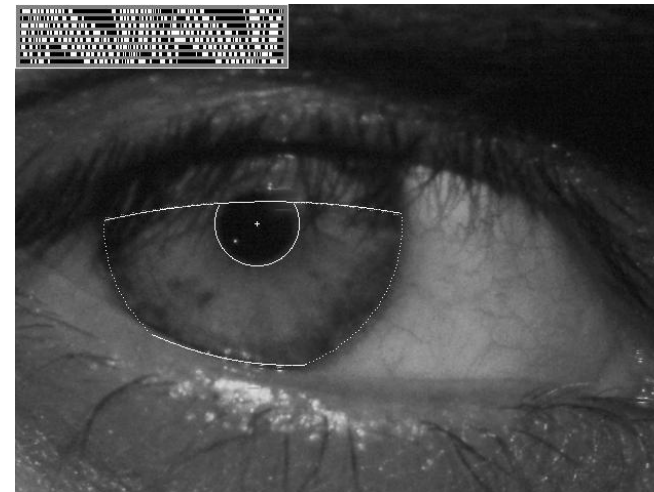
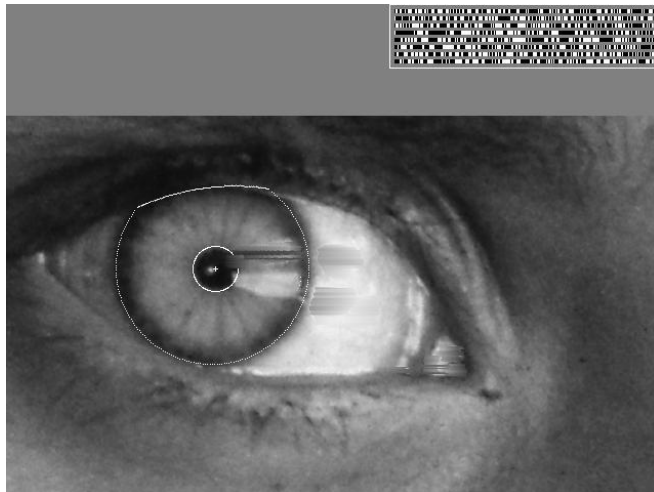
Vision-based Biometrics



Vision-based Biometrics



“How the Afghan Girl was Identified by Her Iris Patterns” Read the [story](#)



Beauty Camera

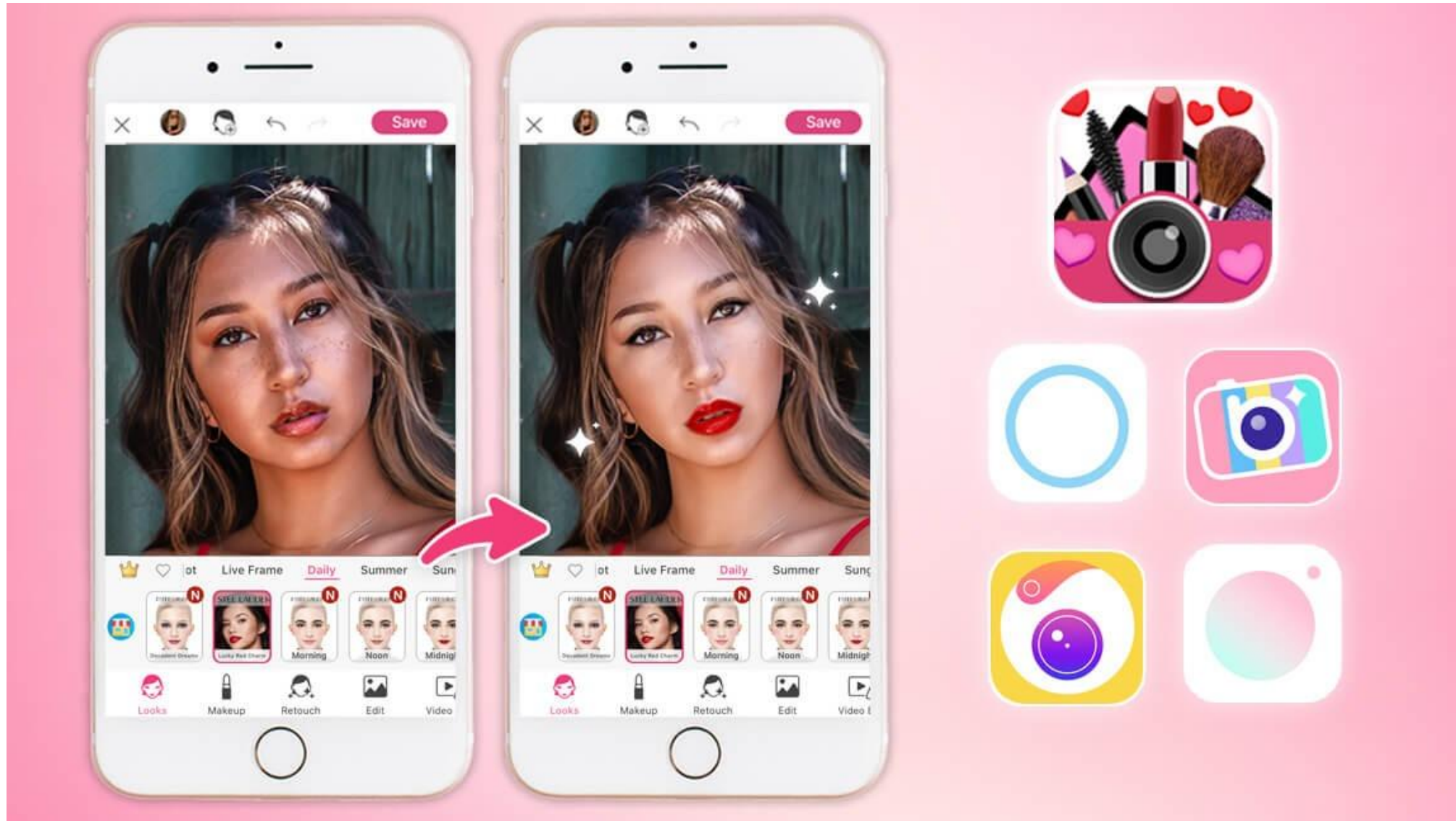


Image Synthesis



Which person is real?

Click on the person who is real.



Image Synthesis

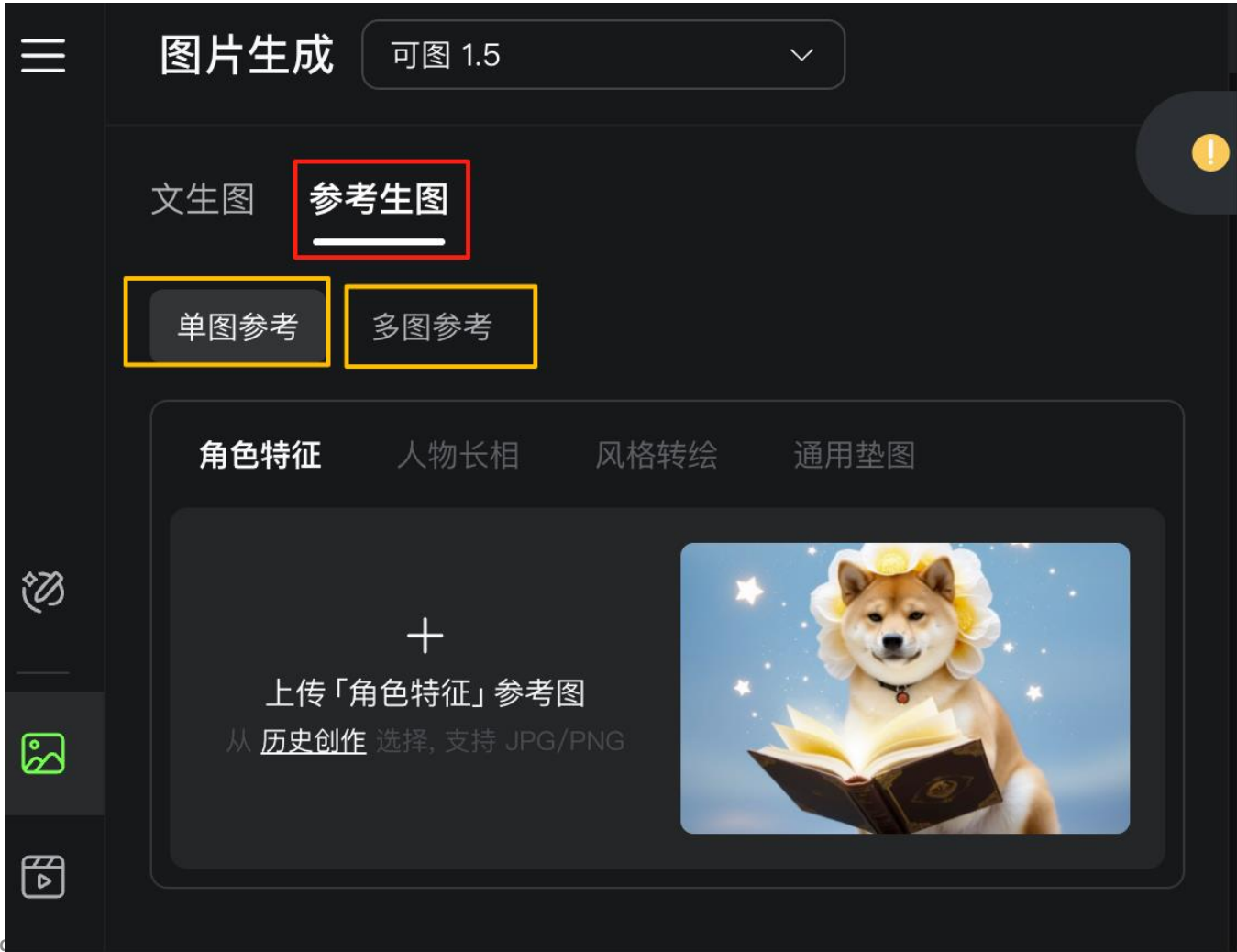


“An astronaut riding a horse in a photorealistic style” – DALL-E 2



“A photo of a Corgi dog riding a bike in Times Square. It is wearing sunglasses and a beach hat” – Imagen

Personalization Image Generation



Personalization Image Generation



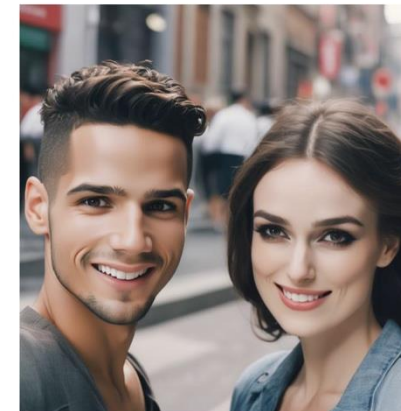
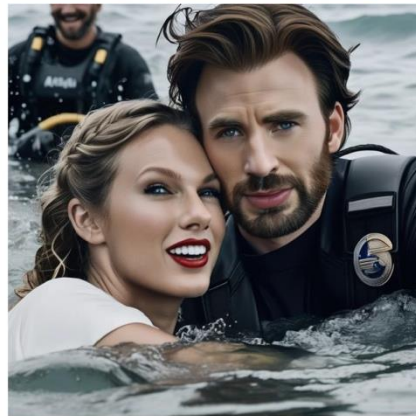
Personalization Image Generation

Reference

OMG (LoRA)

Reference

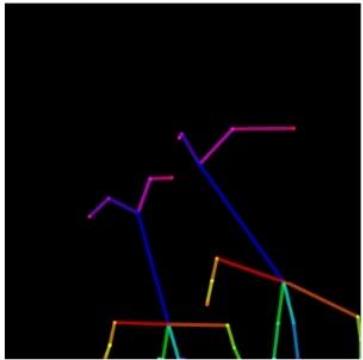
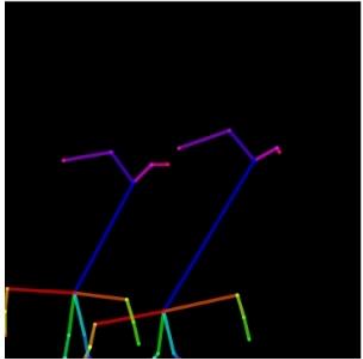
OMG (InstantID)



Personalization Image Generation



a **man** and a **woman** on the street



a) ControlNet with Human Pose

Personalization Image Generation

a **man** and a **woman** on the beach



b) ControlNet with Canny Edge

Personalization Image Generation

a **man** and a **woman** in the restaurant



c) ControlNet with Depth

Personalization Image Generation

Realistic style

Anime Sketch style

Comic Book Style

Cyberpunk Style

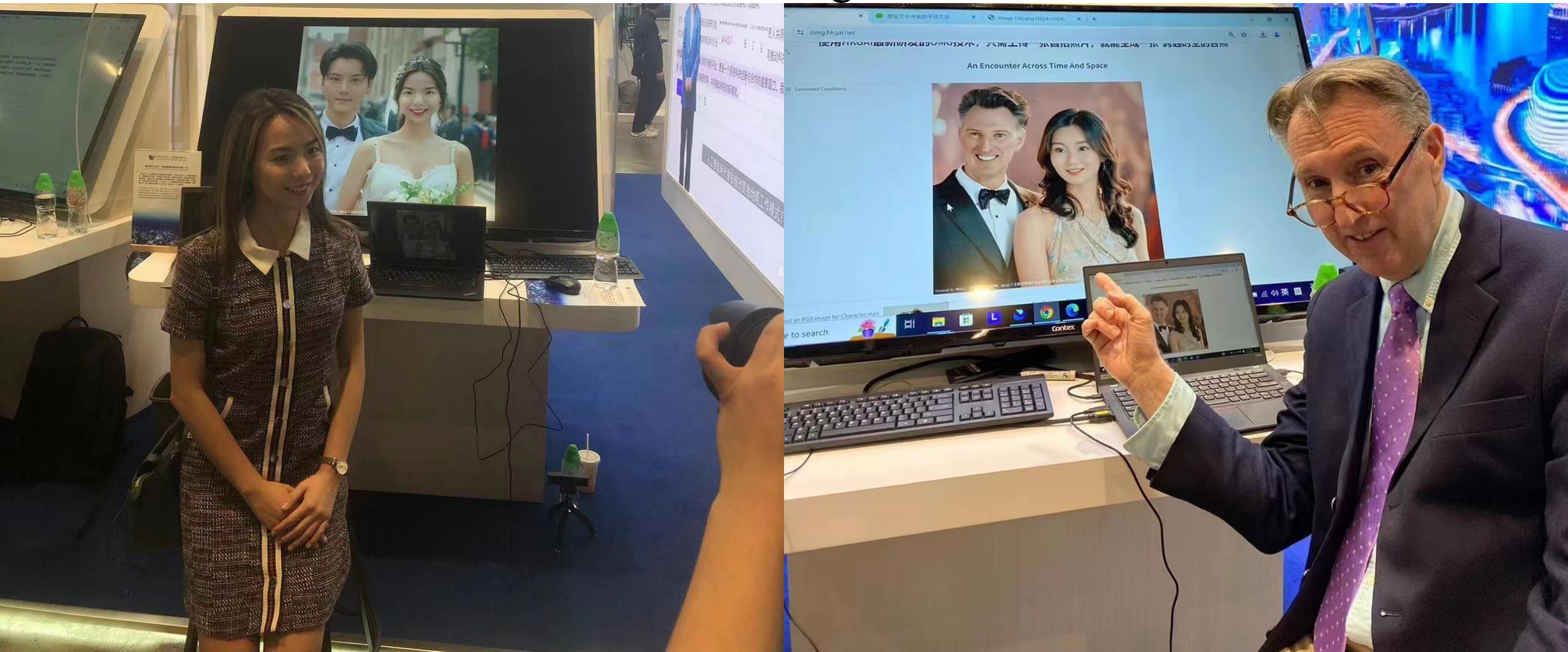
OMG
(LoRA)



OMG
(InstantID)



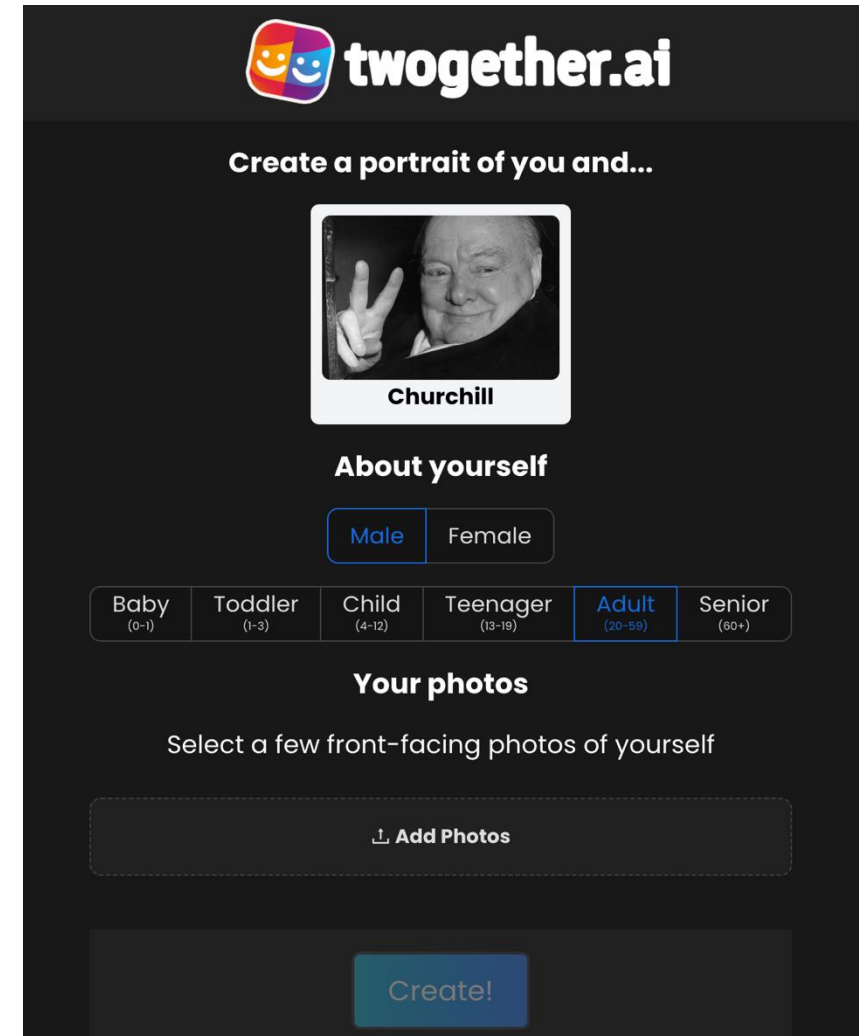
Personalization Image Generation



Personalization Image Generation

**OMG: Occusion-friendly Personalized
Multi-concept Generation in Diffusion Models**

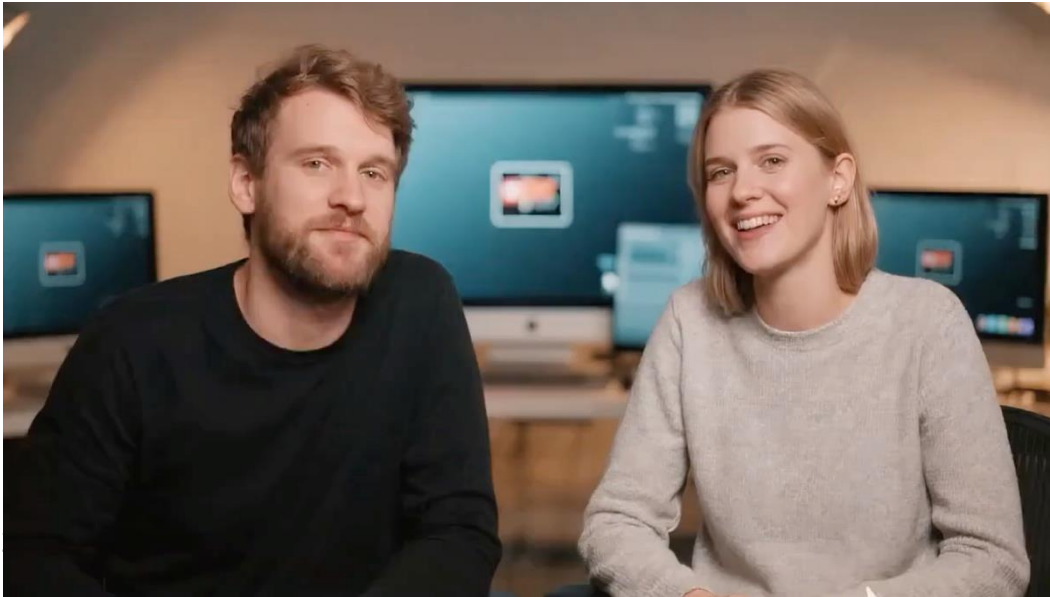
Personalization Image Generation



Talking Face



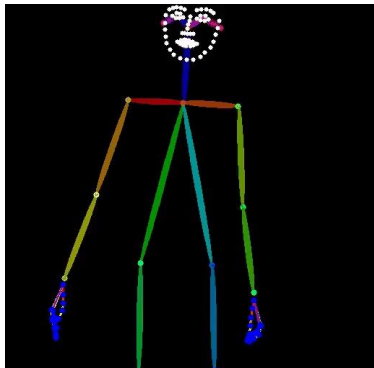
Talking Face



Talking Face (scale to more person)



Human Body Animation



Human Body Animation



Personalization Video Generation



Personalization Video Generation



Video Style Transfer



Stylized Video Generation (with different prompts)



Stylized Video Generation (with different styles)



Video Super Resolution

DAM-VSR: Disentanglement of Appearance and Motion for Video Super-Resolution

Video Editing

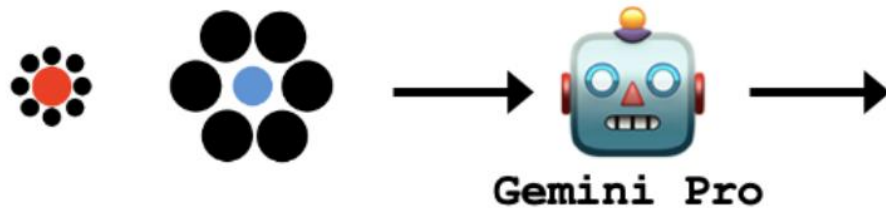


3D Generation (Material)

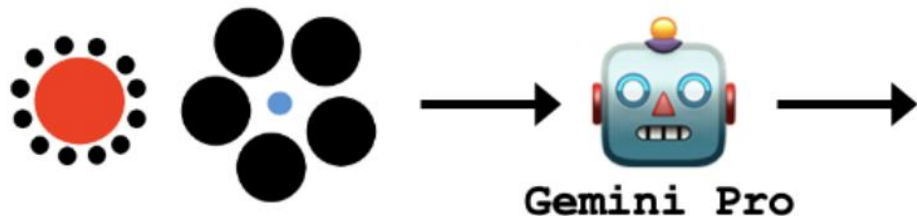


Current models still makes silly mistakes

Prompt: "Which is bigger, the blue circle or the red circle?"



"This is a classic optical illusion! The red and blue circles are actually the same size. Our brains tend to perceive the red circle as smaller because it's surrounded by larger black circles. The blue circle, surrounded by smaller black circles, appears larger in comparison."



"This is a classic optical illusion! The red and blue circles are actually the same size. Our brains tend to perceive the red circle as larger because it's surrounded by smaller circles, while the blue circle is surrounded by larger circles. This is an example of the Ebbinghaus illusion."

[Tomer Ullmann, *The Illusion-Illusion: Vision Language Models See Illusions Where There are None*, arXiv 2024]

Why is computer vision difficult?



Viewpoint variation



Illumination



Credit: Flickr user mich

Scale

Why is computer vision difficult?



Intra-class variation



Motion (Source: S. Lazebnik)

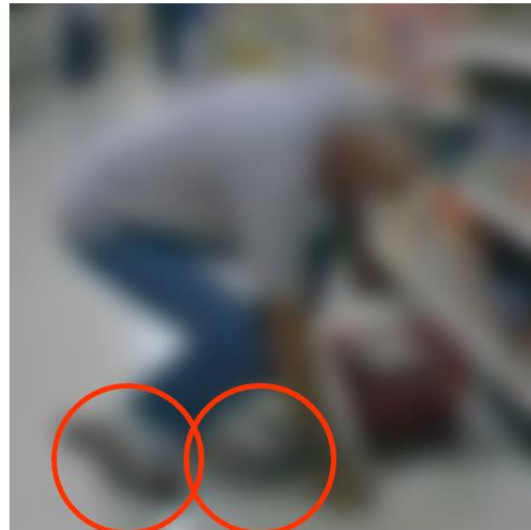
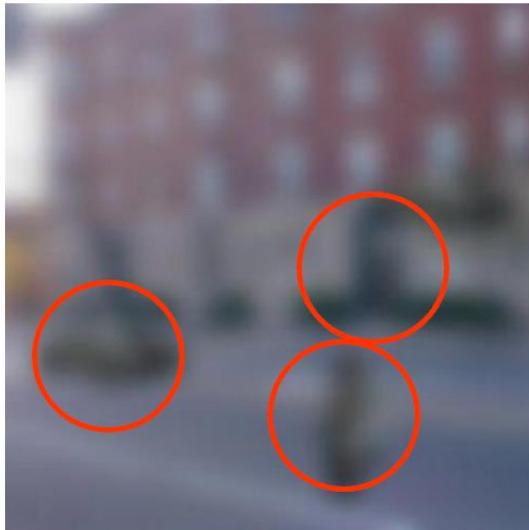
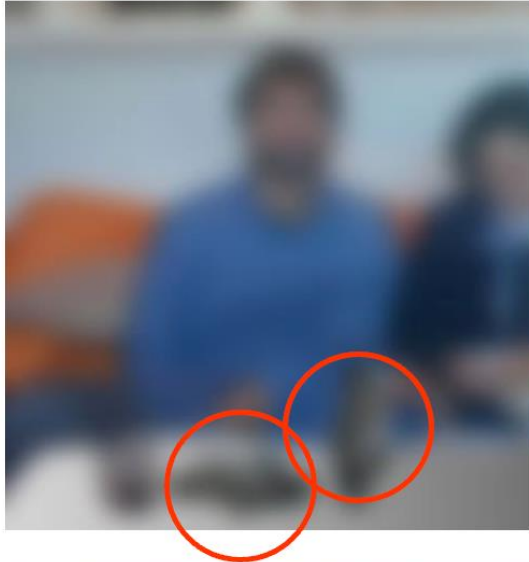


Background clutter



Occlusion

Why is computer vision difficult?



Local ambiguity

Introduction to Computer Vision (CV)

- What is Computer Vision
- Brief History of Computer Vision
- Key Research Areas of Computer Vision
- Applications of Computer Vision
- Why Study Computer Vision

Why Study Computer Vision

Computer Vision is ~~almost~~ everywhere

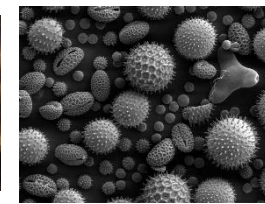
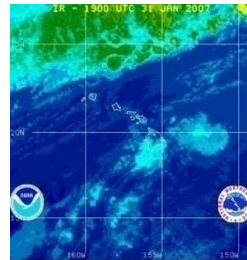
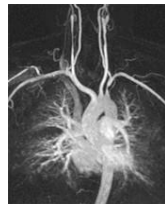
Why Study Computer Vision



flickr

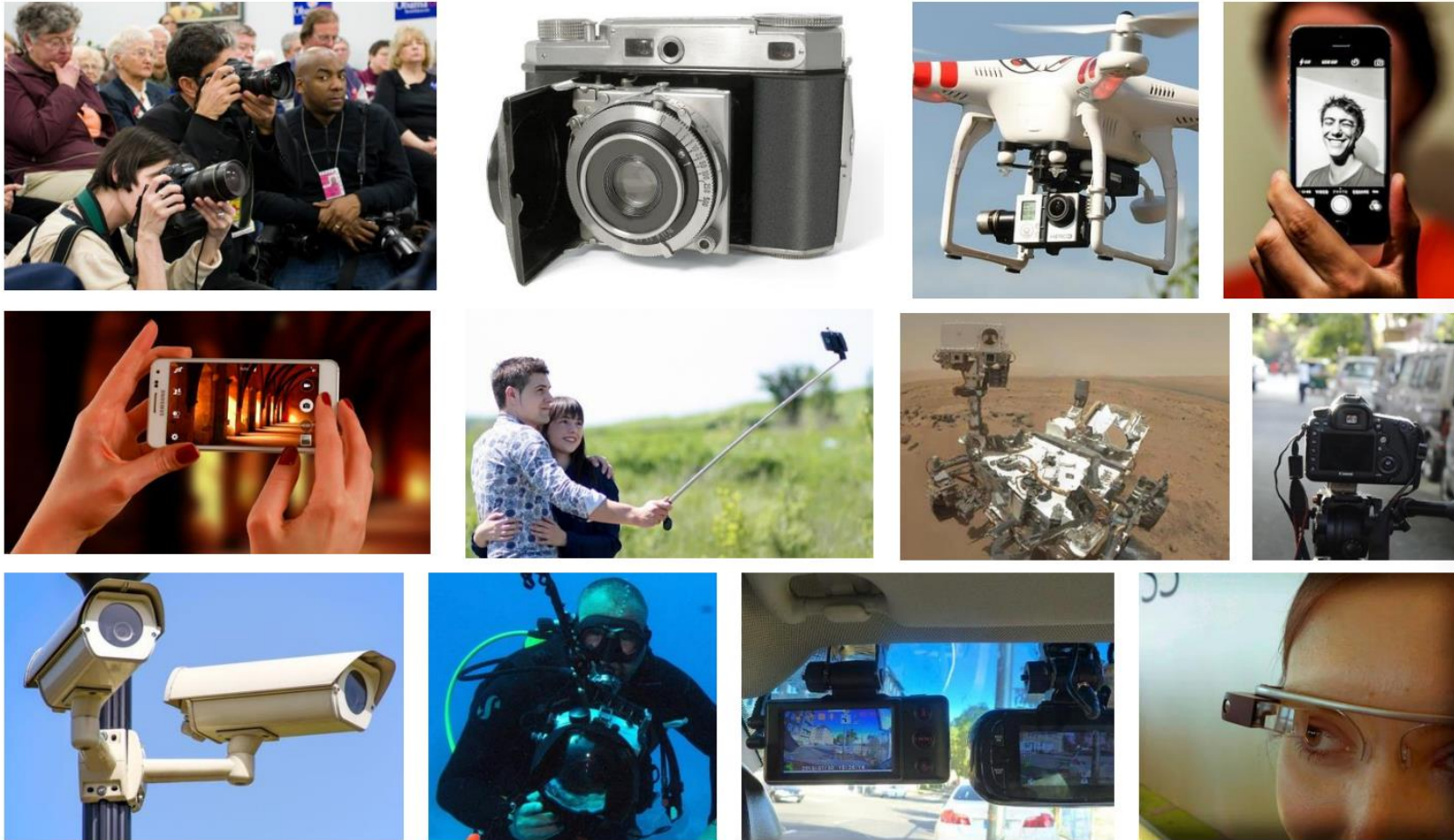


YouTube



Billions of images/videos captured per day

Why Study Computer Vision



Why Study Computer Vision

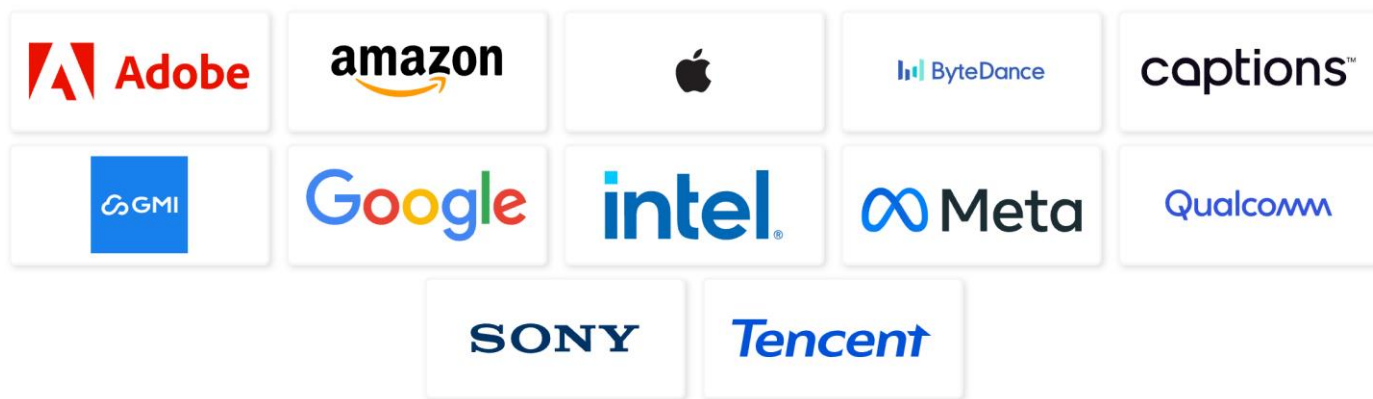
It is a good time to do computer vision

Why Study Computer Vision



Why Study Computer Vision

PLATINUM



PLATINUM



Gold

Ultimate



Google DeepMind



Platinum



amazon | science



Silver



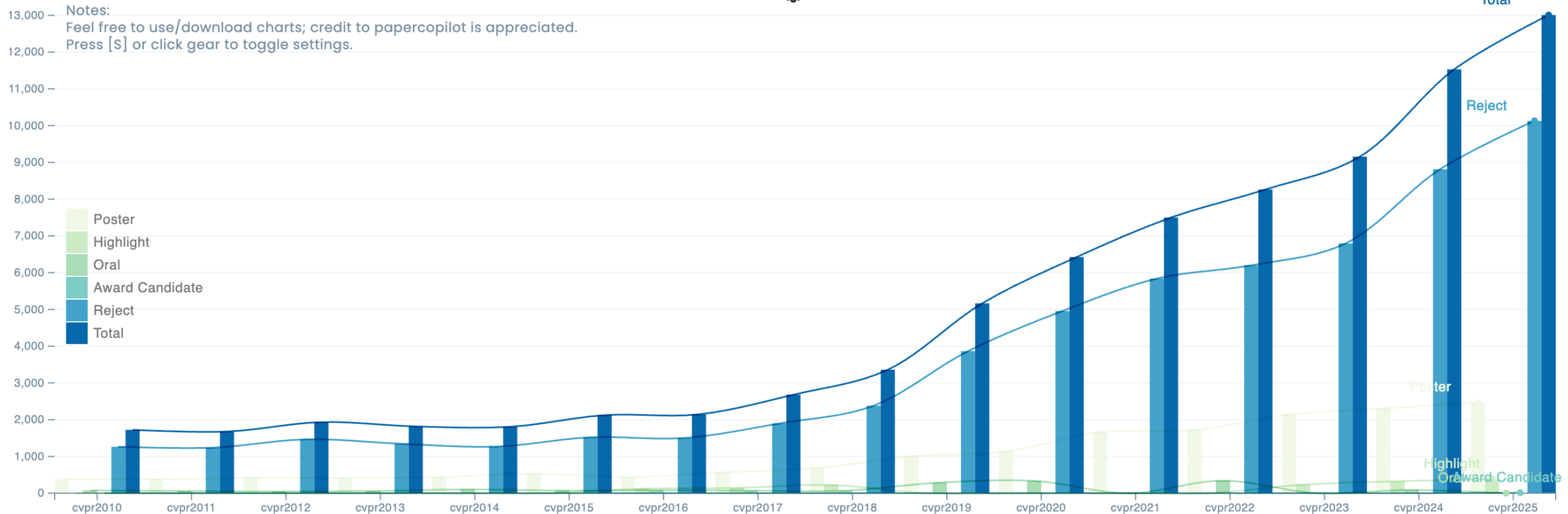
Google Research



Industry aggressively hiring CV graduates, or even students (dominant industrial presence at top CV conference like CVPR/ICCV for recruitment)

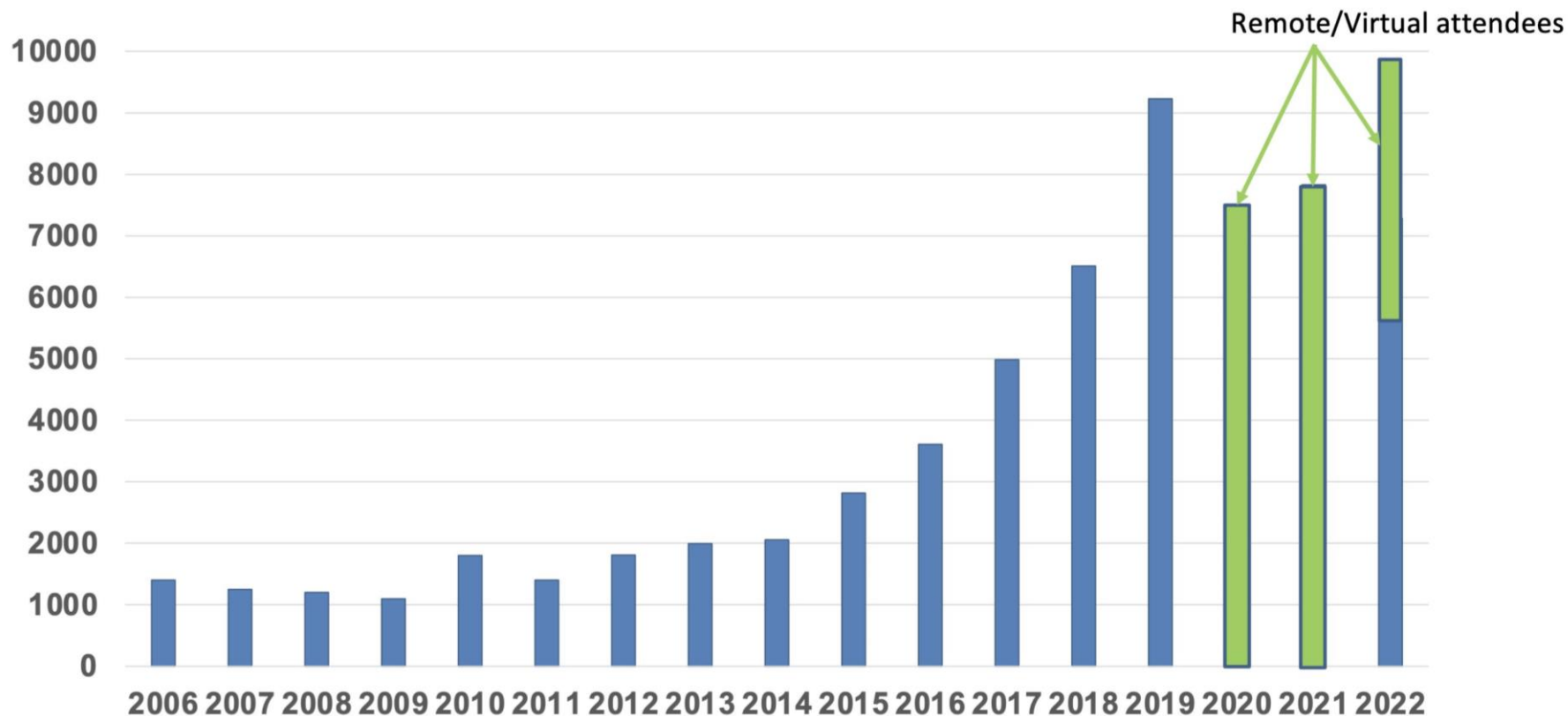
Why Study Computer Vision

↑ Frequency



Why Study Computer Vision

CVPR Attendance Trend (as of June 20, 2022)



Why Study Computer Vision

Google Scholar

Top publications

Categories > Engineering & Computer Science > Subcategories ▾

	Publication	<u>h5-index</u>	<u>h5-median</u>
1.	IEEE/CVF Conference on Computer Vision and Pattern Recognition	<u>440</u>	689
2.	Neural Information Processing Systems	<u>337</u>	614
3.	Advanced Materials	<u>327</u>	420
4.	International Conference on Learning Representations	<u>304</u>	584
5.	IEEE/CVF International Conference on Computer Vision	<u>291</u>	484
6.	Journal of Cleaner Production	<u>272</u>	359
7.	International Conference on Machine Learning	<u>268</u>	424

Q & A